
A Unified Convergence Theory for Non-Stationary Reinforcement Learning

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Abstract

Non-stationary reinforcement learning lacks a unified convergence theory: variation-budget regret bounds, sliding-window methods, tempo-and-forgetting analyses, two-term decompositions, and causal-RL frameworks each address fragments without composing. We present a unified theory through the composition of four structural elements: (i) a two-gap diagnostic separating goal-feasibility from policy-quality; (ii) a point-mass reverse-KL/TV identity under deterministic optimum, $\text{TV}(\delta_{a^*}, Q) = 1 - e^{-D_{\text{KL}}}$, an *exact* algebraic identity that lies strictly below the Pinsker and Bretagnolle–Huber envelopes at this deterministic corner (collapsing the BH inequality’s envelope to a single value rather than introducing a new bounding technique); (iii) a multi-factor strategic tempo $\mathcal{T}_\Sigma = \min_{(i,j)} \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij})$ with a structural forgetting prerequisite $\overline{\mathcal{T}}_\Sigma > \rho_\Sigma / R_\Sigma$ as a survival inequality; (iv) closed-loop interventional access making regret bounds learnable from on-policy data via the Pearl causal hierarchy. Composing these gives a cumulative dynamic regret theorem with rate $O(V_{\max} \sqrt{(B_T + 1)T})$ where B_T counts optimum-change events. We further establish a structural failure class: every gain-decay update (effective per-element gain $\alpha_{ij}^{(t)} \rightarrow 0$, including count-accumulating Bayesian schemes and Robbins–Monro gradient methods) eventually violates the forgetting prerequisite for any positive disturbance rate, while finite-gain mechanisms admit bidirectional thresholds (constant-step stochastic approximation, sliding windows, bounded memory, block restart). The point-mass identity is coordinate-optimal among bounds depending only on total variation, chain-rule compositional to multi-step trajectory regret as behavior-cloning loss against the optimal trajectory, and load-bearing for the cumulative theorem. The deterministic-optimum scope extends perturbatively to ϵ -stochastic and softmax-regularized policies with $O(\epsilon \log(1/\epsilon))$ and $O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$ corrections respectively (exponentially small with polynomial prefactor). Theory-only; Lemma 5.2 makes the ProST reduction rigorous at the sector-parameter level, while their published experiments motivate the strategic-tempo component.

1 Introduction

Non-stationary reinforcement learning has matured along four largely separate research tracks. Each has produced rigorous results; none has been composed with the others into a single convergence theory.

The four tracks are visible in the recent literature as parallel lineages. **Variation-budget dynamic regret under drift**^[1–4] gives sublinear dynamic regret under bounded total variation in reward and transition dynamics. **Two-term regret decompositions**^[5–7] split the dynamic regret along an *exploration-vs-adaptation* axis, isolating the cost of confidence-set construction from the cost of

tracking a moving optimum. **Tempo and forgetting analyses**^[8–12] make policy-update timing or evidence-discount rate an explicit convergence variable. **Causal and interventional access**^[13–18] uses causal-graph structure to sharpen sample complexity, but operates in stationary settings only.

The information-theoretic regret literature^[19–24] traverses these tracks at a different layer, using entropy, mutual information, information ratio, Pinsker, or Hellinger to bound regret. Notably absent from that literature, as we document below (Appendix F), is an *exact* reverse-KL / total-variation identity at the deterministic-optimum corner: a sharp action-distribution regret coordinate that strictly improves both Pinsker and the Bretagnolle–Huber inequality^[25] at this point. The Bretagnolle–Huber inequality is itself absent from the retrieved RL/non-stationary corpus *as an upper-bound regret coordinate* (its use in lower-bound proofs via Le Cam’s method^[26,27] is well established and not contested here); the *exact identity at its point-mass corner* — which we establish — is *a fortiori* absent.

These four tracks share an evident common ancestor — the dynamic-regret analysis of online MDPs — but no published framework composes them. In particular, no framework combines (a) a regret decomposition along the *goal-feasibility-vs-policy-quality* axis (distinct from the exploration-vs-adaptation decompositions); (b) an exact point-mass reverse-KL/TV regret identity under deterministic optimum, strictly tighter than the Bretagnolle–Huber inequality at this corner; (c) a structural survival inequality threading rate-of-policy-revision against environment-side disturbance; and (d) a closed-loop causal-access argument that makes the regret bound *learnable* on-policy rather than merely provable analytically. The fundamental question remains open:

Can we design a unified convergence theory for non-stationary reinforcement learning that handles non-stationarity, has explicit metric structure on policy space, and is learnable from on-policy data?

In this paper we present the first composition that delivers all three properties jointly. Composing four structural elements — a two-gap diagnostic, a point-mass reverse-KL/TV identity, a multi-factor strategic tempo with forgetting prerequisite, and closed-loop interventional access — yields a cumulative dynamic regret theorem with rate $\text{DynReg}(T) \leq \tilde{O}(N_h \sqrt{(B_T + 1)T})$ in the piecewise-stationary regime, where B_T counts piecewise-stationary segment boundaries (kernel-and-reward changes) along the realized trajectory and N_h is the planning horizon. The per-round coordinate of this rate is sharper than both Pinsker and Bretagnolle–Huber at the deterministic- π^* corner. The technical anchor is an exact algebraic identity: $\text{TV}(\delta_{a^*}, Q) = 1 - e^{-D_{\text{KL}}(\delta_{a^*} \| Q)}$, which holds for any policy Q with $Q(a^*) > 0$ and lies *strictly below* both the Pinsker envelope $\sqrt{D_{\text{KL}}/2}$ and the Bretagnolle–Huber envelope $\sqrt{1 - e^{-D_{\text{KL}}}}$ at this corner.

Roadmap Section 2 situates the four-track landscape. Section 3 sets minimal preliminaries. Section 4 presents the four-component composition theorem with narrative unpacking and a defense of why each component is necessary. Section 5 sketches the proof via four key lemmas chained through a block decomposition. Section 6 gives concluding observations and practitioner takeaways. Full proofs and auxiliary mathematical material are in the appendices.

1.1 Contribution

The contribution has the shape — recognized by the NeurIPS Theory Track guidelines as a valid form of originality — of “*a novel combination of existing techniques [where] the reasoning behind this combination is well-articulated.*” Three specific moves carry the paper.

(i) Point-mass reverse-KL/TV regret identity (Section 4, full development Section 5.1). Under deterministic optimum policy $\pi^* = \delta_{a^*}$, an elementary calculation gives the *exact identity* $D_{\text{KL}}(\pi^* \| Q) = -\log Q(a^*) = -\log(1 - \text{TV}(\pi^*, Q))$, equivalently $\text{TV}(\pi^*, Q) = 1 - e^{-D_{\text{KL}}(\pi^* \| Q)}$. Substituting this into the classical Bretagnolle–Huber inequality $\text{TV}(P, Q) \leq \sqrt{1 - e^{-D_{\text{KL}}(P \| Q)}}$ yields $1 - e^{-D_{\text{KL}}} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$, which holds *strictly* on $D_{\text{KL}} > 0$ since $x < \sqrt{x}$ on $(0, 1)$. At the deterministic- π^* corner the identity supplies the *exact* TV value strictly below the BH envelope, positioning BH as the lineage the identity sits within rather than as the source it instantiates. Composed with the textbook total-variation regret bound (full development Section 5.1), the identity yields a tight two-sided characterization $\Delta_{\min}(1 - e^{-D_{\text{KL}}(\pi^* \| Q)}) \leq R(Q) \leq V_{\max}(1 - e^{-D_{\text{KL}}(\pi^* \| Q)})$, *strictly below* the Pinsker form $V_{\max} \sqrt{D_{\text{KL}}/2}$ — at $D_{\text{KL}} = 0.01$ the identity bound is $7\times$ tighter; at

88 $D_{\text{KL}} > 2$ Pinsker becomes vacuous against the trivial V_{max} envelope while the identity continues to
 89 saturate at V_{max} . The deterministic- π^* scope is not a hard wall: the identity extends *perturbatively* to
 90 ϵ -greedy and softmax-regularized optima with quantified $O(\epsilon \log(1/\epsilon))$ and $O(\tau^{-1} \exp(-\Delta_{\text{min}}/\tau))$
 91 corrections respectively (full development in Section 5.1 and Appendix B.2), recovering exact equality
 92 at $\epsilon = 0 / \tau \rightarrow 0$.

93 **(ii) Strategic tempo with forgetting prerequisite (Section 4, full development Section 5.2).** A
 94 multi-factor strategic tempo $\mathcal{T}_{\Sigma}$ — composed per revisable policy element of an observation rate
 95 ν , an identifiability coefficient ι , and an update gain η — yields a structural survival inequality
 96 $\mathcal{T}_{\Sigma}^{\text{bn,ss}} := \min_{(i,j) \in E} \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij}) > \rho_{\Sigma}/R_{\Sigma}$ in which all three factors are multiplicatively
 97 load-bearing per element, with the *bottleneck* (per-element minimum) as aggregator. The familiar
 98 single-factor form $(1 - \lambda) > \rho_{\Sigma}/R_{\Sigma}$ is recovered as a corollary under uniform-Regime-A Beta-
 99 Bernoulli normalization. A structural-class theorem identifies the *gain-decay* class $\mathcal{A}_{\text{decay}}$ (count-
 100 accumulating Bayesian updates, observation-aggregating schemes without restart, gradient-based
 101 Robbins–Monro decaying-step methods) as one in which every agent eventually violates the threshold
 102 for any positive disturbance rate, while non-accumulating mechanisms (constant-step SA, sliding-
 103 window, bounded-memory, block-restart) face *bidirectional* ceilings. This *lifts* the tempo result of^[8]
 104 from a single-factor hyperparameter-optimization claim to a multi-factor structural-threshold claim.
 105 The forgetting prerequisite is the strategic analog of the persistence condition in adaptive control^[28],
 106 with $(1 - \lambda)$ playing the role of adaptive tempo per element.

107 **(iii) Closed-loop interventional access (Section 4, full development Section 5.3).** With explicit
 108 grounding in Bareinboim’s causal-hierarchy theorem^[29], the agent in the feedback loop *generates*
 109 Pearl Level-2 (interventional) data by construction: the action a_t causally precedes o_{t+1} , so under
 110 positivity, sequential ignorability, and a known action mechanism, the loop’s data-generating process
 111 is interventional rather than associational. This makes the regret bound of (i) *learnable* from on-policy
 112 interaction in regimes with sufficient identifiability, not merely provable in principle. The argument is
 113 implicit in the action-perception-loop framing of active inference^[30,31] and the causal-RL line^[13–18];
 114 we make the Bareinboim-hierarchy connection explicit, partition usable strength of identification
 115 into three regimes (A: full; B: partial; C: observation-only), and flag honestly where the loop yields
 116 interventional *data* without yielding identified *do*-estimates.

117 **Connective tissue: the two-gap diagnostic (Section 4).** A separate decomposition of the regret
 118 signal — the *satisfaction gap* δ_{sat} (the goal is unmet under the best available one-step improvement)
 119 versus the *control regret* δ_{regret} (the gap between the best available improvement and the agent’s current
 120 policy) — provides the corrective-action routing that ties (i)–(iii) together. The 2×2 disambiguation
 121 routes four regimes (success, strategy problem, capability limit, both) to four distinct corrective
 122 actions, along an axis orthogonal to the exploration-vs-adaptation decompositions of^[5–7].

123 **Composition** The four together yield (Section 4) a non-stationarity-aware convergence theory with
 124 three properties no existing strand exhibits jointly: the bound *handles non-stationarity* (via tempo and
 125 forgetting); has *explicit metric structure on policy space* (via the point-mass reverse-KL identity); is
 126 *learnable from on-policy data* (via closed-loop interventional access). Composition with a variation-
 127 budget block argument under a restart-on-change base learner yields the cumulative dynamic regret
 128 bound stated above — sharper per-round coordinate than Pinsker/BH, with the cumulative rate sitting
 129 in the piecewise-stationary B_T family of^[1] (*not* directly comparable to the continuous-variation
 130 rate $\tilde{O}(SA V_T^{1/3} H T^{2/3})$ of^[3]; the two regimes coincide up to constants under abrupt-magnitude
 131 piecewise-stationarity). Each of the four components is itself a theorem (cited or proved); the paper’s
 132 contribution is the recognition that the four together close a story none of the strands closes individually,
 133 plus the cumulative-regret rate that follows from composing the per-round identity with a base-learner
 134 stationary rate.

135 1.2 Scope and limitations

136 **What “convergence” means here** We do not claim pointwise policy convergence — under genuine
 137 non-stationarity the optimum is itself moving. By *convergence* we mean: per-round regret-coordinate
 138 convergence within stationary blocks; cumulative dynamic regret with sublinear / vanishing-average
 139 rate when $B_T = o(T)$ (Cesàro tracking $\frac{1}{T} \sum_t (V_t^* - V(\pi_t)) \rightarrow 0$); and ultimate boundedness of
 140 the strategic mismatch $\|\delta_{\Sigma}\|$ under the tempo threshold. Pointwise $V(\pi_t) \rightarrow V^*$ is structurally
 141 unavailable under genuine non-stationarity; the Cesàro tracking statement is the right replacement.

Theory only No experiments. The point-mass identity is mathematically airtight (a two-line direct calculation that specializes the BH family at its deterministic- π^* corner). The empirical grounding for the strategic-tempo claim is provided indirectly through reduction to^[8] ProST as a special case (Section 5); at the steady-state sector-parameter level the reduction is rigorous.

Canonical scope The composition theorem holds under deterministic optimum policy, bounded value range, isolated optimum (so $\Delta_{\min} > 0$), and a singular causal trajectory in the sense made precise in Section 3. The deterministic- π^* scope extends *perturbatively* to ϵ -stochastic and softmax-regularized optima (Section 5.1); tied-optimum is a degraded one-sided form (Appendix C.5). Each axis of the canonical scope degrades cleanly outside it (we discuss the failure modes in Section 6).

Comparator regime The cumulative rate is in the piecewise-stationary B_T family, not the continuous-variation V_T family of^[3]. Generalizing the per-round-identity route to continuous variation is open (Section 6).

2 Related Work

The four tracks named in Section 1 organize the literature.

Variation-budget dynamic regret ^[1–4,32] establish dynamic regret bounds under a *variation budget* V_T on cumulative reward and transition variation.^[3] achieves the near-optimal continuous-variation rate $\tilde{O}(SA V_T^{1/3} H T^{2/3})$ via RestartQ-UCB;^[1] treats the piecewise-stationary case with restart-on-change. Our cumulative rate $\tilde{O}(N_h \sqrt{(B_T + 1)T})$ sits in the piecewise-stationary family — same block-shape as Cheung-style restart-on-change — recovered as a corollary of composition with the per-round identity. The novelty is the *threshold form* of the strategic-tempo inequality (environment regimes where no $\lambda \in (0, 1)$ stabilizes the sector-Lyapunov mismatch), which dynamic-regret-optimization analyses do not surface.

Two-term regret decompositions along the exploration-vs-adaptation axis ^[5–7] decompose the dynamic regret into a *confidence-set construction* term and an *adaptation-under-non-stationarity* term.^[33] treats a structurally adjacent question — constraint-region feasibility — using a two-term decomposition along that axis. Same shape, different axis: ours is goal-feasibility-vs-policy-quality (the satisfaction gap distinguishes “the goal is unattainable from M_t ” from “the current policy is suboptimal”); theirs is uncertainty-source. Neither subsumes the other: a Long-Fei Li-style exploration term can be present in any of our 2×2 cells, and an adaptation term likewise. The two axes are orthogonal.

Tempo and forgetting ^[8] (ProST) defines an agent tempo as the schedule of policy update times $\{t_1, \dots, t_K\}$ and computes the schedule that minimizes a dynamic regret upper bound.^[9] (Pausing Policy Learning) refines this with non-zero hold durations.^[10–12] establish forgetting / sliding-window analyses for non-stationary linear MDPs and bandits. We *lift* the ProST move along two axes simultaneously: single-factor $\rightarrow$ multi-factor (per-element $\nu \cdot \iota \cdot (1 - \lambda)$ rather than scalar update frequency); and hyperparameter-optimization $\rightarrow$ structural-survival inequality (the threshold below which no schedule persists, irrespective of which schedule is chosen). ProST recovers as the Beta-Bernoulli special case $|E| = 1, \nu = \iota = 1$.

Causal and interventional access ^[13–18,34] use causal-graph structure to sharpen sample complexity in RL. Their setting is *stationary*. We compose the closed-loop interventional-access argument with non-stationarity. The connection between Pearl’s causal hierarchy^[35] and the data-generating process of an agent in a feedback loop^[29] is the technical bridge.

Information-theoretic regret (cross-cutting) ^[19–24,36] traverse the above tracks at a different layer, using entropy / mutual information / information ratio / Pinsker / Hellinger as the divergence backbone. The Russo–Van Roy information ratio uses Pinsker as its Cauchy–Schwarz follow-up; the IDS line and its descendants use mutual-information bounds. The Bretagnolle–Huber inequality^[25] — well-known in lower-bound proofs via Le Cam’s method^[26,27] — does not appear *as an upper-bound regret coordinate* in this corpus, and its exact point-mass corner (our Lemma 5.1) is *a fortiori* absent. We document the retrieval scope and methodology in Appendix F.

191 **Adjacent (satisficing / feasibility)** ^[37,38] use *satisficing* to mean “any policy above acceptance
 192 level β is acceptable.” Our δ_{sat} is structurally distinct: $V_O^{\min}$ is a property of the objective (set by
 193 the application domain), and $\delta_{\text{sat}} > 0$ signals goal-relative *infeasibility* (no policy in Π meets the
 194 threshold), not policy-relative satisficing.

195 **Contemporaneous (post-March 2026)** ^[38,39]; cited and distinguished without empirical-
 196 comparison demand per the contemporaneous-work cutoff convention.

197 3 Preliminaries

198 We adopt strict minimalism: only the notation and assumptions used directly in the main result and
 199 the mechanism statements appear here. Convention-hierarchy details (one-step vs. receding-horizon
 200 vs. Bellman continuation), the action-gap matching argument, tied-optimum extensions, and the
 201 deterministic-trajectory deployment modes are deferred to Appendix C.

202 **Episodic non-stationary MDP** A finite-horizon non-stationary MDP $(\mathcal{S}, \mathcal{A}, P_t, r_t, N_h)$ with state
 203 space $\mathcal{S}$, finite action space $\mathcal{A}$, time-indexed transition kernels $P_t(\cdot \mid s, a)$ and reward functions
 204 $r_t(s, a)$ allowed to vary across rounds t , and finite planning horizon N_h . Standard non-stationary-RL
 205 boundedness assumptions^[1]: per-step reward in $[0, 1]$.

206 **Variation regimes** A variation budget V_T measures continuous variation in transitions and rewards.
 207 We work in the *piecewise-stationary* specialization: time partitions into $B_T + 1$ maximal stationary
 208 intervals on which (P_t, r_t) is constant, with $B_T := |\{t : (P_t, r_t) \neq (P_{t-1}, r_{t-1})\}|$ counting *kernel-*
 209 *stationary-segment boundaries*. Optimum-changes $a_t^* \neq a_{t-1}^*$ can only occur at segment boundaries
 210 (since a^* is determined by (P_t, r_t)); the converse is not generally true (a segment boundary need not
 211 change a^* — e.g., uniform reward shifts that lift all actions, or kernel shifts that preserve the argmax).
 212 Counting kernel-segment boundaries rather than optimum-change events is the right measure for the
 213 base learner’s per-block stationary-regret guarantee in (A5): UCRL2 / UCBVI / Thompson sampling
 214 build empirical confidence sets on (P, r) , and within-block kernel drift breaks the per-block $\sqrt{L}$ rate
 215 even when a^* is unchanged. In the bandit case with isolated optimum and reward shifts that always
 216 cross argmax, B_T reduces to the optimum-change count $|\{t : a_t^* \neq a_{t-1}^*\}|$ — a useful corollary in
 217 regime ARG (Section 4 *Comparator regime*) but not the headline framing. The two variation regimes
 218 (B_T vs. continuous V_T) are distinct in general; under abrupt-magnitude- δ piecewise-stationarity they
 219 coincide up to constants ($V_T \asymp \delta B_T$). Our results are stated for B_T ; the continuous- V_T extension is
 220 open (Section 6).

221 **Policy distribution** The agent’s policy at round t is a distribution $Q_t(\cdot \mid s)$ over actions; we write Q
 222 when state and round are understood. We adopt the canonical scope: the optimum policy at (M_t, s)
 223 is *deterministic*, $\pi^*(\cdot \mid M_t, s) = \delta_{a^*(s)}$ where $a^*(s) := \arg \max_a Q^\pi(s, a)$ is the optimal action
 224 under the agent’s current model M_t . Deterministic π^* is canonical for finite-MDP RL with isolated
 225 optima^[27]; the perturbative extension to ϵ -stochastic and softmax-regularized optima (Section 5.1,
 226 full derivation Appendix B.2) covers smooth deviations.

227 **Value object** Given an objective O , model M_t , policy π , and horizon N_h : $Q_O(M_t, a; \pi_{\text{cont}}, N_h) =$
 228 $\mathbb{E}[V_O(\tau) \mid M_t, \text{do}(a_t = a), a_{t+1} \sim \pi_{\text{cont}}]$. The $\text{do}(\cdot)$ ^[35] flags intervention on a_t , not conditioning.
 229 We adopt the **one-step improvement** convention $\pi_{\text{cont}} = \pi_{\text{current}}$ as default: most conservative, compa-
 230 rable across rounds, fixed-point-free. Receding-horizon and Bellman alternatives are in Appendix C.1.

231 **Bounded value range** $V_{\max}(M_t) := \max_a Q_O(M_t, a) - \min_a Q_O(M_t, a)$, finite under bounded
 232 reward and finite horizon.

233 **Action gap (isolated optimum)** $\Delta(a) := Q_O(a^*) - Q_O(a) \in [0, V_{\max}]$, $\Delta_{\min} := \min_{a \neq a^*} \Delta(a) >$
 234 0 whenever the optimum is isolated.

235 **Strategic regret** For policy distribution Q : $R(Q) := Q_O(a^*) - \mathbb{E}_{a \sim Q}[Q_O(a)] = \sum_{a \neq a^*} Q(a) \cdot$
 236 $\Delta(a)$. Three classical forms — $\mathbb{E}_{\pi^*}[V] - \mathbb{E}_Q[V]$, $V(a^*) - \mathbb{E}_Q[V]$, $\mathbb{E}_{\pi^*}[V - V_Q]$ — coincide under
 237 deterministic π^* .

238 **Total variation and reverse-KL** For finite action measures P, Q on $\mathcal{A}$, $\text{TV}(P, Q) :=$
239 $\frac{1}{2} \sum_a |P(a) - Q(a)|$, $D_{\text{KL}}(P \| Q) := \sum_a P(a) \log \frac{P(a)}{Q(a)}$ with the convention $0 \log 0 = 0$.
240 Our bounds use the *reverse* KL direction (with $P = \pi^*$); the forward direction $D_{\text{KL}}(Q \| \pi^*)$ is $+\infty$
241 whenever Q has off-optimum mass and is therefore vacuous as a regret coordinate (full direction-
242 forcing argument in Appendix C.3).

243 **Singular trajectory** The agent operates on a single, non-forkable causal trajectory: a_t causally
244 precedes o_{t+1} , and the model state M_t ’s sufficiency is defined relative to *this* trajectory rather than to
245 a model-state equivalence class. This is the operational ground for the do-operator above and for the
246 closed-loop interventional-access argument (Section 5.3).

247 4 Main Result

248 We present the four components first as motivation, then state the formal composition theorem, then
249 unpack what its conclusions mean.

250 4.1 The four components

251 Each component supplies a *distinct epistemic property* that the others cannot deliver. The composition
252 is necessary for the joint properties (rate + persistence + learnability + corrective-action routing), not
253 for the rate alone. The component-by-component failure-mode analysis is in Section 4.4 below.

254 **Component 1: Two-gap diagnostic — corrective-action routing** Let Π denote the policy class
255 available to the agent and let $V_O^{\min}$ denote the threshold trajectory value above which the objective is
256 “met.” Define *objective attainability* $A_O(M_t; \Pi, N_h) := \sup_{\pi \in \Pi} V_O(M_t, \pi; N_h)$, the *satisfaction*
257 *gap* $\delta_{\text{sat}} := V_O^{\min} - A_O$ (goal-feasibility diagnostic; positive means *unmet* under the best available
258 policy), and the *control regret* $\delta_{\text{regret}} := A_O - V_O(M_t, \pi_{\text{current}}; N_h) \geq 0$ (policy-quality diagnostic).
259 The 2×2 cross of these two gaps routes corrective action: *Success* (no action), *Strategy problem*
260 (revise the policy), *Capability limit* (revise the model, policy class, or horizon — in that order), *Both*
261 (revise the policy first, then re-evaluate). Without this routing, an agent receiving a regret signal
262 cannot distinguish “the goal is unattainable from M_t ” from “the current policy is suboptimal.” A
263 single $\delta_{\text{objective}} = V_O^{\min} - V_O(M_t, \pi_{\text{current}}; N_h)$ collapses both regimes into a single number and loses
264 the diagnostic. The two-gap split runs along an axis *orthogonal* to the exploration-vs-adaptation
265 decompositions of^[5–7]: theirs isolates uncertainty *sources*; ours isolates *what is wrong*.

266 **Component 2: Point-mass reverse-KL/TV identity — metric layer on policy space** Under
267 deterministic $\pi^* = \delta_{a^*}$, the reverse-KL and total variation between π^* and any policy Q are related by
268 the *exact identity* $D_{\text{KL}}(\pi^* \| Q) = -\log Q(a^*) = -\log(1 - \text{TV}(\pi^*, Q)) \iff \text{TV}(\pi^*, Q) =$
269 $1 - e^{-D_{\text{KL}}(\pi^* \| Q)}$. This is a two-line direct calculation under the point-mass assumption (full statement
270 and proof in Section 5.1). It is *not* the Bretagnolle–Huber inequality at equality: substituting it into
271 the BH form $\text{TV} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$ gives $1 - e^{-D_{\text{KL}}} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$, *strictly* on $D_{\text{KL}} > 0$. The
272 identity sits *strictly below* the BH envelope at this corner. Composed with the textbook TV-regret
273 bound $R(Q) \leq V_{\max} \text{TV}(\pi^*, Q)$ and the matching action-gap lower bound, it yields the two-sided
274 characterization $\Delta_{\min}(1 - e^{-D_{\text{KL}}(\pi^* \| Q)}) \leq R(Q) \leq V_{\max}(1 - e^{-D_{\text{KL}}(\pi^* \| Q)})$, *Lipschitz-*
275 *equivalent* with constants $\Delta_{\min}/V_{\max}$ and 1. The identity coordinate is *coordinate-optimal among*
276 *bounds depending only on TV*: any tighter bound on $R(Q)$ requires information beyond TV (the
277 value-landscape spread). Strict improvement over Pinsker is uniform: at $D_{\text{KL}} = 0.01$ the identity
278 bound is $7 \times$ tighter; for $D_{\text{KL}} > 2$ Pinsker is vacuous against the trivial $V_{\max}$ envelope while the identity
279 saturates at $V_{\max}$. The deterministic- π^* scope is not a hard wall: the identity extends *perturbatively* to
280 ϵ -greedy and softmax-regularized optima with corrections $O(\epsilon \log(1/\epsilon))$ and $O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$
281 respectively — exponentially small with polynomial prefactor in the softmax case (Section 5.1).

282 **Component 3: Multi-factor strategic tempo with forgetting prerequisite — non-stationarity**
283 **persistence** Consider a structured policy with $|E|$ revisable elements (in our analysis, edges of
284 an internal causal model the agent maintains over policy components; in Lee et al. 2023^[8]’s ProST,
285 sub-policies indexed by a tempo schedule). For each element (i, j) , let ν_{ij} denote the per-element
286 observation rate, $\iota_{ij} \in [0, 1]$ the identifiability coefficient (fraction of evidence that genuinely identifies

the causal effect rather than reflecting confounded association), and $\eta_{\text{edge},ij}$ the per-element update gain. Within a sector-Lyapunov reduction with diagonal correction (full Model (Σ) in Section 5.2), per-element exponential forgetting at rate λ_{ij} gives steady-state gain $\eta_{\text{edge},ij}^{\text{ss}} \approx 1 - \lambda_{ij}$, and ultimate boundedness of the strategic mismatch $\|\delta_\Sigma\|$ within the modeled dynamics is *sufficient* under the bottleneck threshold $\mathcal{T}_\Sigma^{\text{bn,ss}} := \min_{(i,j) \in E} \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij}) > \rho_\Sigma / R_\Sigma$. All three factors are multiplicatively load-bearing per element (zeroing any collapses the bottleneck; low identifiability cannot be compensated by faster forgetting), and the *bottleneck* (per-element minimum, not aggregate sum) is the right aggregator: a Lyapunov derivation under adversarial disturbance saturates at min, not $\sum$, when the disturbance concentrates on the weakest element. The structural-class theorem identifies the *gain-decay* class $\mathcal{A}_{\text{decay}}$ — count-accumulating Bayesian updates without forgetting (e.g., Beta-Bernoulli with $\eta = 1/(n+1)$), bounded-memory schemes with growing memory, observation-aggregating schemes without restart, *and* gradient-based methods with vanishing step sizes (the Robbins–Monro regime, $\eta_t = c/t^\beta$ for $\beta \in (0, 1]$) — as one in which *every* agent eventually violates the threshold for any positive disturbance rate (formal statement in Appendix C.6). Non-accumulating mechanisms (constant-step SA, sliding-window, bounded-memory, block-restart) face *bidirectional* ceilings: the threshold is tight in both directions and the schedule has to be in the right regime, not arbitrarily fast or slow. The familiar single-factor form $(1 - \lambda) > \rho_\Sigma / R_\Sigma$ is recovered as a corollary under uniform-Regime-A Beta-Bernoulli normalization.

Component 4: Closed-loop interventional access — on-policy learnability Pearl’s causal hierarchy^[35] and the causal-hierarchy theorem of^[29] establish that Level-2 (interventional) queries $P(Y \mid \text{do}(X))$ are not in general identifiable from Level-1 (associational) data $P(Y \mid X)$. By temporal ordering and the singular-trajectory commitment of Section 3, a_t causally precedes o_{t+1} . Under three conditions on the agent’s action mechanism — (C1) *positivity* ($\pi_t(a \mid H_t) \geq p_{\min} > 0$ on the support of the information state H_t), (C2) *sequential ignorability* (the information state blocks unobserved confounding of action selection), and (C3) *known action mechanism* (the behavior policy is known) — conditioning on H_t blocks the action-selection confounding pathway, so the conditional distribution $P(o_{t+1} \mid a_t, H_t)$ coincides with the interventional distribution $P(o_{t+1} \mid \text{do}(a_t), H_t)$. The loop’s data-generating process is *interventional* by construction (formal statement in Section 5.3). This makes the regret bound of Component 2 *learnable on-policy* in regimes with sufficient identifiability: the agent’s identified optimum $a_{\text{ag},t}^*$ matches the true optimum $\bar{a}_t^*$ with probability p_{id} , and the per-state, per-step bias from misidentification is bounded by $(1 - p_{\text{id}}(s)) \log(1/q_0)$ under two-point support $Q_t \geq q_0$ (Section 5.4). Three regimes^[29] partition usable strength: A (intervention-rich, $\iota \approx 1$), B (partial, $\iota \in (0, 1)$), C (observation-only, $\iota \approx 0$).

4.2 The composition theorem

Theorem 4.1 (Composition: joint guarantees with conditional cumulative reduction). *Let $(\mathcal{S}, \mathcal{A}, P_t, r_t, N_h)$ be a non-stationary MDP with bounded reward, finite action space, and for each round t a deterministic optimum policy $\pi_t^* = \delta_{a_t^*}$ with action gap $\Delta_{\min} > 0$. Suppose the agent operates on a singular causal trajectory in the sense of Section 3 and updates its policy with per-element exponential forgetting at rates $\{\lambda_{ij}\}$. Let $K_t(s) := D_{\text{KL}}(\delta_{a_t^*(s)} \parallel Q_t(\cdot \mid s))$ be the per-state reverse-KL coordinate (in the bandit case $N_h = 1$ the state argument is suppressed). If*

(A1) Metric extension. The identity $\text{TV}(\delta_{a_t^(s)}, Q_t(\cdot \mid s)) = 1 - e^{-K_t(s)}$ of Lemma 5.1 is read in the extended real sense: when $Q_t(a_t^*(s) \mid s) = 0$, $K_t(s) = +\infty$ and $1 - e^{-K_t(s)} = 1$ recovers the trivial bound $R(Q_t \mid s) \leq V_{\max}$. No pointwise positivity is required; vanilla deterministic UCB / UCBVI / UCRL2 deployed at the optimism-argmax are in scope without modification.*

(A2) Multi-factor forgetting prerequisite. Within Model (Σ) of Section 5.2, the bottleneck strategic tempo $\mathcal{T}_\Sigma^{\text{bn,ss}} := \min_{(i,j) \in E} \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij}) > \rho_\Sigma / R_\Sigma$.

(A3) Identifiability regime. The action mechanism is known and (C1)–(C3) of Section 5.3 hold; the agent operates in Regime A or Regime B with per-state argmax-correctness probability $p_{\text{id}}(s)$, with floor $p_{\text{id}} := \min_s p_{\text{id}}(s)$.

(A4) Two-gap diagnostic. The agent applies the satisfaction-gap / control-regret 2×2 to route corrective action.

(A5) Piecewise-stationary block structure with restart-on-change identity-tight base learner. The time axis partitions into $B_T + 1$ stationary intervals on which the MDP is stationary and $\pi_t^(\cdot \mid s) =$*

340 $\delta_{a_t^*}(s)$ is fixed per state. The base learner is restarted at each block boundary; within each restarted in-
 341 terval of length L it satisfies $\sum_{t=1}^L \mathbb{E}[\overline{\text{TV}}_t] \leq 2c\sqrt{L}$, where $\overline{\text{TV}}_t := \frac{1}{N_h} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}} [\text{TV}(\pi_t^*(\cdot \mid$
 342 $s_h), Q_t(\cdot \mid s_h))]$ is the occupancy-weighted per-round coordinate. (A non-restarting carryover
 343 variant is plausible but requires bounding cross-block occupancy shift; see Remark in Section 5.)

344 Then:

345 **(i) Per-round regret is two-sided identity-bounded.** At every visited state, $\Delta_{\min}(1 - e^{-K_t(s)}) \leq$
 346 $R(Q_t \mid s) \leq V_{\max}(1 - e^{-K_t(s)})$.

347 **(ii) Aggregate strategic mismatch is ultimately bounded.** Under (A2), the modeled mismatch $\|\delta_{\Sigma}\|$ is
 348 ultimately bounded within $R_{\Sigma}^* = \rho_{\Sigma} / \mathcal{T}_{\Sigma}^{\text{bn,ss}}$.

349 **(iii) The KL coordinate is computable with controlled bias.** $-\log Q_t(a_{\text{ag},t}^*(s) \mid s)$ is computable
 350 directly from Q_t . Per-state bias: $\mathbb{E}[|\hat{D}_t(s) - D_t^{\text{true}}(s)|] \leq (1 - p_{\text{id}}(s)) \log(1/q_0)$ under $Q_t(\cdot \mid s) \geq q_0$
 351 at the two argmax candidates. Vanishes in Regime A.

352 **(iv) The 2×2 cell identifies the corrective-action class.** $(\delta_{\text{sat}}, \delta_{\text{regret}})$ routes to revise-policy / revise-
 353 model-class-or-horizon / revise-objective.

354 **(v) Trajectory-level blockwise dynamic regret.** Under (A5), $\mathbb{E}[\text{DynReg}(T)] :=$
 355 $\mathbb{E}\left[\sum_{t=1}^T (V_t^{\pi_t^*} - V_t^{Q_t})\right] \leq 2c V_{\max} N_h \sqrt{(B_T + 1)T} + V_{\max} N_h (1 - p_{\text{id}}) \cdot T$.

356 The proof composes the per-round identity (i) with the simulation lemma^[40–43], a block decomposition
 357 at optimum-change events, Cauchy–Schwarz across the $B_T + 1$ blocks, and the bias bound from (iii).
 358 Full proof in Appendix A; the four key lemmas are surfaced in Section 5.

359 4.3 Unpacking the conclusions

360 Theorem 4.1 asserts that the four components compose into a non-stationarity-aware convergence
 361 theory with three properties that no existing strand has individually: it *handles non-stationarity*
 362 (via (ii)/(A2)), has *explicit metric structure on policy space* (via (i)/Component 2), and is *learnable*
 363 *on-policy* (via (iii)/(A3)). Conclusion (iv) is the connective tissue routing corrective action; conclusion
 364 (v) is the cumulative regret rate that follows from composing the per-round metric coordinate with the
 365 base-learner stationary rate.

366 We emphasize four things about this theorem.

367 *The per-round coordinate is exact, not an inequality.* Under the canonical scope (deterministic π^*), the
 368 per-round identity-bound (i) is $\Delta_{\min}/V_{\max}$ -Lipschitz-equivalent to the regret. Both endpoints of the
 369 Lipschitz envelope are *achieved* on extremal value landscapes ($\Delta(a) = V_{\max}$ for all $a \neq a^*$ saturates
 370 the upper bound; $\Delta_{\min} = \max_{a \neq a^*} \Delta(a)$ saturates the lower). The per-round coordinate is therefore
 371 *coordinate-optimal among bounds depending only on TV*: tightening would require value-landscape
 372 information beyond TV. This is sharper than Pinsker’s $\sqrt{D_{\text{KL}}/2}$ — exact instead of approximate —
 373 and sharper than the Bretagnolle–Huber envelope $\sqrt{1 - e^{-D_{\text{KL}}}}$ at this corner since $x < \sqrt{x}$ on $(0, 1)$.

374 *The cumulative rate is in the piecewise-stationary B_T family; the continuous-variation V_T family is*
 375 *reachable via Wei-Luo MASTER black-box wrapping.* The lifted rate $\tilde{O}(N_h^2 \sqrt{SA(B_T + 1)T})$ that
 376 follows under UCRL2/UCBVI base learners (Appendix D) sits in the same block-shape family as the
 377 restart-on-change variants of^[11] — the route here is through the per-round identity rather than direct opti-
 378 mism. It is *not directly comparable per-form* to the continuous-variation rate $\tilde{O}((SA)^{1/3} V_T^{1/3} H T^{2/3})$
 379 of^[3]: the parameter (B_T vs. V_T), the T -scaling ($T^{1/2}$ vs. $T^{2/3}$), the S, A -scaling ($\sqrt{SA}$ vs. $(SA)^{1/3}$),
 380 and the horizon dependence (N_h^2 vs. N_h) all differ. Under abrupt-magnitude- δ piecewise-stationarity
 381 the two regimes coincide up to constants ($V_T \asymp \delta B_T$), and B-CS1’s rate is sharper for fixed B_T ;
 382 outside that, Mao-style continuous-variation analysis is sharper for fixed V_T . The per-round identity
 383 route extends to continuous variation via^[2]’s MASTER black-box reduction, achieving Best-of-
 384 Both-Worlds dynamic regret $\tilde{O}(\min\{\sqrt{(B_T + 1)T}, (V_T + 1)^{1/3} T^{2/3}\})$ automatically without prior
 385 knowledge of either budget; full development in Section 6 (*On the continuous-variation extension*)
 386 and theorem-grade availability via the (A5’)-BoBW Remark in Appendix A.

387 *The bias term in (v) is the value-coordinate misidentification penalty.* The second term $V_{\max} N_h (1 -$
388 $p_{\text{id}}) \cdot T$ is the per-round expected value gap between the true and agent-identified optima — bounded
389 by $V_{\max}$ on the misidentification event of probability $1 - p_{\text{id}}$, aggregated over the N_h horizon steps. It
390 vanishes in Regime A ($p_{\text{id}} \rightarrow 1$ under (C1)–(C3)); it is linear in T — vacuous as a rate but bounded by
391 the trivial value envelope $V_{\max} N_h \cdot T$ — outside Regime A. In Regime B ($p_{\text{id}} \in (0, 1)$), the dominant
392 rate term is the metric one provided $1 - p_{\text{id}}$ is small enough relative to $\sqrt{(B_T + 1)/T}$. In Regime C
393 ($p_{\text{id}} \rightarrow 0$) the bound saturates at the trivial envelope; algorithm flags this regime as out-of-scope for
394 on-policy learnability (Appendix D). *Strict assumption-set weakening:* the rate (v) does *not* require
395 Lemma 4’s two-point support condition $Q_t(\cdot | s) \geq q_0$ — that condition is needed only for conclusion
396 (iii)’s diagnostic computability of the agent’s KL coordinate, not for the cumulative-regret rate itself.

397 *The horizon factor is the standard simulation-lemma penalty.* The N_h in conclusion (v) is the linear-
398 in- N_h per-round simulation-lemma coefficient — sharper than the N_h^2 of TRPO-style worst-state
399 bounds^[44]. The lifted N_h^2 in the UCRL2/UCBVI cumulative rate reflects the additional N_h in the
400 within-block base-learner constant c .

401 4.4 Necessity of the four components

402 A natural reviewer question: why exactly four? The composition is a *bundle of compatible guarantees*
403 (each conclusion supported by its own assumptions), not a single integrated theorem in which every
404 component is necessary for the rate. The rate (v) goes through (A5) and (i) alone; the joint properties
405 — rate plus persistence plus learnability plus corrective-action routing — require all four.

406 **Without Component 2 (point-mass identity):** the per-round bound is Pinsker $V_{\max} \sqrt{D_{\text{KL}}/2}$ or BH
407 $V_{\max} \sqrt{1 - e^{-D_{\text{KL}}}}$ — both strictly looser at this corner; Pinsker is vacuous for $D_{\text{KL}} > 2$. The metric
408 layer loses exactness, and the “behavior-cloning loss against optimal trajectory” interpretation of the
409 cumulative coordinate vanishes.

410 **Without Component 3 (strategic tempo):** the agent has no persistence guarantee. By the structural-
411 class theorem on $\mathcal{A}_{\text{decay}}$, every count-accumulating update without forgetting eventually violates
412 the threshold for any positive disturbance; the modeled mismatch $\|\delta_{\Sigma}\|$ grows unboundedly. The
413 per-round bound (i) still holds, but $K_t(s)$ may itself drift — there is no guarantee the per-round
414 coordinate stays bounded.

415 **Without Component 4 (loop-Level-2):** the bound’s RHS is computable from Q_t , but its interpreta-
416 tion as regret-against-true-optimum requires identifying $a_t^*(s)$ — an interventional query under the
417 environment’s transition kernel. In Regime C ($\iota \approx 0$), the bias saturates at $\log(1/q_0)$ per state per
418 step, contributing $N_h \log(1/q_0) \cdot T$ to cumulative regret — the rate (v) becomes vacuous.

419 **Without Component 1 (two-gap diagnostic):** the regret signal δ_{regret} alone does not distinguish
420 “policy revision is the right move” from “the goal is unattainable from M_t .” An agent might persist in
421 policy revision when the model, policy class, or horizon is the actual problem (Capability-limit cell),
422 or revise the goal when policy revision would have sufficed.

423 So while the rate (v) goes through (A5) and (i) alone, the *bundle* — rate plus persistence plus
424 learnability plus corrective-action routing — requires all four. Each drop is a concrete loss of an
425 epistemic property that the field treats as separate.

426 5 Mechanism

427 This section overviews the proof of Theorem 4.1 via four key lemmas chained through a block
428 decomposition. Each lemma is stated formally; intuitions are given in plain English with the algebra
429 deferred to the appendices. The rate (v) is the composition’s load-bearing conclusion; we sketch its
430 derivation in four steps after surfacing the lemmas.

431 5.1 Key Lemma 1: Point-mass reverse-KL/TV identity

432 **Lemma 5.1** (Point-mass reverse-KL/TV identity). *For deterministic $\pi^* = \delta_{a^*}$ (per visited state,*
433 *with $a^* = a^*(s)$ and $Q = Q(\cdot | s)$) and any policy Q , $D_{\text{KL}}(\pi^* \| Q) = -\log Q(a^*) =$
434 $-\log(1 - \text{TV}(\pi^*, Q))$, equivalently $\text{TV}(\pi^*, Q) = 1 - e^{-D_{\text{KL}}(\pi^* \| Q)}$. The identity is read in the*

435 *extended real sense: when $Q(a^*) = 0$ both sides equal 1, with the natural convention $D_{\text{KL}} = +\infty$*
 436 *and $e^{-\infty} = 0$ — the identity holds for all policies Q , including Diracs at suboptimal actions.*

437 *Intuition.* The reverse-KL collapses under the point-mass: $D_{\text{KL}}(\delta_{a^*} \parallel Q) = \sum_a \delta_{a^*}(a) \log \frac{\delta_{a^*}(a)}{Q(a)}$,
 438 where the convention $0 \log 0 = 0$ kills every $a \neq a^*$ term, leaving $\log \frac{1}{Q(a^*)} = -\log Q(a^*)$. Total
 439 variation collapses just as cleanly: $\text{TV}(\delta_{a^*}, Q) = \frac{1}{2} \sum_a |\delta_{a^*}(a) - Q(a)| = 1 - Q(a^*)$. Combining
 440 gives the identity. Substituting into the Bretagnolle–Huber inequality $\text{TV}(P, Q) \leq \sqrt{1 - e^{-D_{\text{KL}}(P \parallel Q)}}$
 441 yields $1 - e^{-D_{\text{KL}}} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$, strictly on $D_{\text{KL}} > 0$. The identity is therefore not BH-at-equality;
 442 it is the *exact value* at the deterministic- π^* corner, sitting strictly below the BH envelope. Full proof
 443 in Appendix B.

444 *Per-round regret bound.* Composed with the textbook TV-regret bound $R(Q) \leq V_{\max} \text{TV}(\pi^*, Q)$
 445 and the matching action-gap lower bound $R(Q) \geq \Delta_{\min} \text{TV}(\pi^*, Q)$, the identity yields the two-sided
 446 characterization that gives Theorem 4.1’s conclusion (i). The bound is *coordinate-optimal among*
 447 *bounds depending only on TV*: any tighter bound on $R(Q)$ requires value-landscape information
 448 beyond TV.

449 *Perturbative extension.* The deterministic- π^* scope is not a hard wall. For ϵ -greedy π_ϵ^* with Q
 450 satisfying full-support $Q(a) \geq q_0$, $\text{TV}(\pi_\epsilon^*, Q) = 1 - e^{-D_{\text{KL}}(\pi_\epsilon^* \parallel Q)} + O(\epsilon \log(1/\epsilon))$ uniformly in
 451 Q ; the proof is a triangle-inequality + expansion argument given in Appendix B.2. For softmax-
 452 regularized $\pi_\tau^* \propto \exp(Q_O/\tau)$, the off-optimum mass concentrates as $O(\exp(-\Delta_{\min}/\tau))$ and the
 453 correction is $O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$ — exponentially small with polynomial prefactor. The two-sided
 454 regret bound transfers with the same correction order. Outside the perturbative regime — genuinely
 455 high-entropy optima, tied-optimum, hard-exploration — the BH inequality becomes the relevant
 456 general bound; Pinsker is the textbook fallback.

457 5.2 Key Lemma 2: Forgetting prerequisite

458 The strategic mismatch state $\delta_\Sigma = (\delta_{ij})_{(i,j) \in E}$ in the Euclidean norm $\|\delta_\Sigma\|^2 = \sum_{(i,j)} \delta_{ij}^2$ evolves
 459 in discrete time as $\mathbb{E}[\Delta \delta_{ij} \mid \delta_{ij}] = -\alpha_{ij} \delta_{ij} + w_{ij}$, $|w_{ij}| \leq \rho_\Sigma / |E|^{1/2}$, where $\alpha_{ij} \geq 0$ is
 460 a per-element correction rate (the strategic sector parameter), w_{ij} is a bounded disturbance, and
 461 the disturbance budget satisfies $\sum_{(i,j)} w_{ij}^2 \leq \rho_\Sigma^2$. The strategic reserve R_Σ is the operative norm-
 462 radius beyond which the agent loses tracking. Beta-Bernoulli edge updates instantiate this *Model* (Σ)
 463 approximately; gradient-based decaying-step updates instantiate it under standard Robbins–Monro
 464 conditions; we will refer to it as *Model* (Σ) throughout.

465 **Lemma 5.2** (Multi-factor forgetting prerequisite). *Within Model (Σ) with diagonal per-element*
 466 *correction rates $\alpha_{ij} = \nu_{ij} \cdot \iota_{ij} \cdot \eta_{\text{edge},ij}$ (per-element observation rate $\nu_{ij} \in [0, 1]$, regime-adjusted*
 467 *identifiability $\iota_{ij} \in [0, 1]$, edge-update gain $\eta_{\text{edge},ij} > 0$), under per-element exponential forgetting*
 468 *at rate $\lambda_{ij} \in (0, 1)$ giving steady-state $\eta_{\text{edge},ij}^{\text{ss}} \approx 1 - \lambda_{ij}$, ultimate boundedness of $\|\delta_\Sigma\|$ within*
 469 *$R_\Sigma^* = \rho_\Sigma / \mathcal{T}_\Sigma^{\text{bn,ss}}$ is sufficient under $\mathcal{T}_\Sigma^{\text{bn,ss}} := \min_{(i,j) \in E} (\nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij})) > \rho_\Sigma / R_\Sigma$. The*
 470 *threshold is sharp inside the diagonal sector model: when the inequality reverses, an adversarial*
 471 *disturbance concentrating on the bottleneck element drives $\|\delta_\Sigma\|$ beyond R_Σ within the modeled*
 472 *dynamics.*

473 *Intuition (why bottleneck, not sum).* The Lyapunov sector product is $\delta_\Sigma^\top \mathbf{F}(\delta_\Sigma) = \sum_{(i,j)} \alpha_{ij} \delta_{ij}^2$.
 474 Bounding this from below uniformly in δ_Σ requires the largest constant c such that $\sum_{(i,j)} \alpha_{ij} \delta_{ij}^2 \geq$
 475 $c \cdot \|\delta_\Sigma\|^2$ holds for all δ_Σ . The largest such c is $\min_{(i,j)} \alpha_{ij}$, not $\sum_{(i,j)} \alpha_{ij}$ — adversarial concentration
 476 of the disturbance on the *weakest* element saturates the bound at the bottleneck. The same effect
 477 makes per-dimension persistence conditions sharper than scalar conditions in adaptive control^[45]:
 478 scalar tempo can overstate the available margin because the worst-case disturbance concentrates on
 479 the slowest-correcting dimension. Ultimate boundedness then follows from Khalil’s classical sector-
 480 Lyapunov template applied to the strategic substate. Full proof in Appendix B; the structural-class
 481 theorem on $\mathcal{A}_{\text{decay}}$ (gain-decay updates universally violate the threshold) is in Appendix C.6.

482 *ProST as a sector-level reduction.* The block-restart structure of ProST^[8] idealizes as an *impulsive*
 483 *system* — between scheduled update times the policy is held and the mismatch evolves under dis-
 484 turbance alone (continuous-destabilizing in the Lyapunov $V = \|\delta_\Sigma\|^2$); at each update time the

policy is revised, contracting the modeled mismatch by per-update impulse gain $\gamma \in (0, 1]$. The reverse-average-dwell-time framework of^[46] (Theorem 1, reverse-ADT branch) gives the threshold $\Delta_{\max} \cdot \rho_{\Sigma}/R_{\Sigma} < -\ln(1 - \gamma)^2$ where $\Delta_{\max}$ is the longest inter-update gap. Under uniform blocks $\Delta_i = T/K$ this reduces to $K/T > \rho_{\Sigma}/(2\gamma R_{\Sigma})$ to leading order in γ — recovering ProST’s K/T form with the impulse gain made explicit. Under nonuniform schedules the impulsive form is *strictly stronger*: the longest block sets the threshold, not the average. This makes a tradeoff between update frequency $1/\Delta_{\max}$ and per-update impulse strength γ visible at the sector-Lyapunov level — the same disturbance can be tolerated either by frequent weak updates or rare strong ones.

5.3 Key Lemma 3: Closed-loop interventional access

Lemma 5.3 (Loop generates interventional samples). *Let H_t denote the agent’s information state at round t (model state M_t , history of observations and actions). Under (C1) positivity — $\pi_t(a \mid H_t) \geq p_{\min} > 0$ on the support of H_t for the actions of interest; (C2) sequential ignorability — H_t d -separates a_t from o_{t+1} in the mutilated graph $G_{\overline{a_t}}$ (equivalently, $a_t \perp\!\!\!\perp Y^{(a_t)} \mid H_t$ in potential-outcome notation); (C3) known action mechanism — the behavior policy $\pi_t(a \mid H_t)$ is known: executed actions are interventions relative to the environment’s transition kernel, and the loop generates samples from $P(o_{t+1} \mid \text{do}(a_t = a), H_t)$ — i.e., Pearl Level-2 interventional kernels conditional on the decision context. The substantive content for the identification claim is (C2); under the singular-trajectory + agent-as-policy architecture of Section 3, (C1) reduces to realized-action positivity (automatic) and (C3) to agent-policy-queryability (architectural). (C1) and (C3) are stated explicitly for forward-compatibility with off-policy / counterfactual extensions.*

Intuition. By temporal ordering and the singular-trajectory commitment of Section 3, a_t causally precedes o_{t+1} . Under (C1)–(C3), conditioning on the information state H_t blocks the action-selection confounding pathway, so the conditional distribution $P(o_{t+1} \mid a_t, H_t)$ coincides with the interventional distribution $P(o_{t+1} \mid \text{do}(a_t), H_t)$ on the support where $\pi_t(a_t \mid H_t) > 0$. The architecture of the agent (Q-learning, transformer, etc.) does not enter: the loop generates conditional-Level-2 samples whenever (C1)–(C3) hold, irrespective of how the agent updates internally. Whether the agent *exploits* this data for Level-2 reasoning is a separate algorithmic question.

Three regimes of identifiability. Three regimes^[29] partition usable strength of identification: - **Regime A (intervention-rich, $\iota \approx 1$)**. Software tests, controlled labs. do-effects identified cleanly; bound realizable on-policy. - **Regime B (partial, $\iota \in (0, 1)$)**. Mixed observation-intervention. Bound holds for the model the agent identifies; bias $\propto 1 - \iota$. - **Regime C (observation-only, $\iota \approx 0$)**. Passive monitoring. Bound provable analytically but not realizable on-policy; observer-on-sub-agent extensions can recover identifiability in subcases (Appendix C.8).

Action-generated data is Level 2 in *character*, but a clean estimate of $P(o \mid \text{do}(a), H_t)$ requires overcoming four typical obstacles: coverage (diverse actions tried), within-step confounding, delay (consequences past $t + 1$), and partial observability. We honor this distinction. Components 2 and 4 are jointly load-bearing: the identity of Section 5.1 gives the analytic regret-KL relationship; the loop-Level-2 access here gives the data substrate from which π^* — and the bound’s RHS — can be empirically estimated under sufficient identifiability. Neither alone makes the bound learnable.

5.4 Key Lemma 4: Bias bound under partial identifiability

Lemma 5.4 (Bias bound). *Assume $Q_t(a \mid s) \geq q_0 > 0$ for both $a = a_{\text{ag},t}^*(s)$ (the agent’s identified optimum) and $a = \tilde{a}_t^*(s)$ (the true optimum). Let $\hat{D}_t(s) := -\log Q_t(a_{\text{ag},t}^*(s) \mid s)$ and $D_t^{\text{true}}(s) := -\log Q_t(\tilde{a}_t^*(s) \mid s)$. Then $|\hat{D}_t(s) - D_t^{\text{true}}(s)| \leq \mathbf{1}[a_{\text{ag},t}^*(s) \neq \tilde{a}_t^*(s)] \cdot \log(1/q_0)$, and taking expectations: $\mathbb{E}[|\hat{D}_t(s) - D_t^{\text{true}}(s)|] \leq (1 - p_{\text{id}}(s)) \log(1/q_0)$, where $p_{\text{id}}(s) := \Pr[a_{\text{ag},t}^*(s) = \tilde{a}_t^*(s)]$.*

Intuition. The argument is an indicator decomposition. On the matching event, the difference is identically zero. On the misidentification event, $\log Q_t(a_{\text{ag},t}^*(s) \mid s)$ and $\log Q_t(\tilde{a}_t^*(s) \mid s)$ both lie in $[\log q_0, 0]$, so their absolute difference is at most $\log(1/q_0)$. The expectation bound follows from $\Pr[\text{misidentification event}] = 1 - p_{\text{id}}(s)$. Bias scales *linearly* in misidentification mass — vanishes in Regime A ($p_{\text{id}} \rightarrow 1$), controlled in Regime B, saturates at $\log(1/q_0)$ in Regime C. The argmax-correctness probability p_{id} relates to the per-edge identifiability ι via the policy DAG structure; in the simple-bandit case $p_{\text{id}} = \iota$. Note: the support condition is *two-point* (only at the two argmax

536 candidates), strictly weaker than the *full-support* condition $Q(a) \geq q_0$ for all a that the perturbative
 537 identity (Section 5.1) requires — F.1 applies whenever the two candidates are positively-supported
 538 regardless of Q_t ’s mass on other actions. Full proof in Appendix B.

539 5.5 Combining the lemmas: cumulative regret in four steps

540 We sketch the proof of conclusion (v) of Theorem 4.1 — the cumulative trajectory-level dynamic
 541 regret rate — by chaining the four key lemmas through a block decomposition.

542 *Step 1 — Per-round bound via the simulation lemma.* For finite-horizon N_h non-stationary MDPs with
 543 bounded reward, the standard performance-difference / simulation lemma^[40–43] gives, for any two
 544 policies π_t^* and Q_t and initial state s_0 , $V_t^{\pi_t^*}(s_0) - V_t^{Q_t}(s_0) \leq V_{\max} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}} [\text{TV}(\pi_t^*(\cdot |$
 545 $s_h), Q_t(\cdot | s_h))]$ $= V_{\max} \cdot N_h \cdot \overline{\text{TV}}_t$, where $d_h^{Q_t}$ is the round- t learner-induced state distribution and
 546 $\overline{\text{TV}}_t$ is the occupancy-weighted TV defined in (A5). By Key Lemma 1 applied per state, $\text{TV}(\pi_t^*(\cdot |$
 547 $s), Q_t(\cdot | s)) = 1 - e^{-K_t(s)}$ — strictly sharper than Pinsker — preserved underneath the occupancy
 548 aggregation. The horizon factor N_h here is the linear-in- N_h simulation-lemma penalty, sharper than
 549 the N_h^2 of TRPO-style worst-state bounds^[44].

550 *Step 2 — Block decomposition at piecewise-stationary segment boundaries.* Partition $[1, T]$ at (P, r) -
 551 change events into $0 = \tau_0 < \tau_1 < \dots < \tau_{B_T} \leq \tau_{B_T+1} = T$ (the kernel-stationary-segment bound-
 552 aries of Section 3). Within each block $[\tau_i, \tau_{i+1})$ of length Δ_i the MDP is stationary by construction.
 553 (A5) — restart-on-change base learner — gives within-block guarantee $\sum_{t \in \text{block}_i} \mathbb{E}[\overline{\text{TV}}_t] \leq 2c\sqrt{\Delta_i}$.

554 *Step 3 — Cauchy–Schwarz across blocks.* Across the $B_T + 1$ blocks, $\sum_i \sqrt{\Delta_i} \leq \sqrt{(B_T + 1)T}$
 555 by Cauchy–Schwarz. Multiplying by $V_{\max}N_h$ closes the rate term: $\sum_t \mathbb{E}[V_{\max}N_h \overline{\text{TV}}_t] \leq$
 556 $2cV_{\max}N_h \sqrt{(B_T + 1)T}$.

557 *Step 4 — Misidentification penalty via simulation lemma at the identified-vs-true-optimum gap.* Steps
 558 1–3 give the rate term against the agent’s *identified* optimum policy $\pi_{\text{ag},t}^* := \delta_{a_{\text{ag},t}^*}$ (which is what
 559 (A5)’s base-learner guarantee delivers); the dynamic regret in (v) compares to the *true* optimum
 560 $\tilde{\pi}_t^* := \delta_{\tilde{a}_t^*}$. Apply the simulation lemma to these two point-mass policies: each per-step bracket
 561 reduces to $Q_h^{\tilde{\pi}_t^*}(s_h, \tilde{a}_t^*) - Q_h^{\pi_{\text{ag},t}^*}(s_h, a_{\text{ag},t}^*) \in [0, V_{\max}] \cdot \mathbf{1}[\text{misid at } s_h]$ (zero on the matching event by
 562 optimality of $\tilde{a}_t^*$, bounded by $V_{\max}$ on the misidentification event). Aggregated over the N_h horizon
 563 steps with floor $p_{\text{id}} := \min_s p_{\text{id}}(s)$, the per-round misidentification penalty is at most $V_{\max}N_h(1 - p_{\text{id}})$,
 564 contributing $V_{\max}N_h(1 - p_{\text{id}}) \cdot T$ to cumulative regret — *in value coordinates* throughout, no detour
 565 through the KL coordinate. (Key Lemma 4’s KL-readout bound is the right tool for conclusion (iii)’s
 566 diagnostic computability; for the value-coordinate penalty here, indicator decomposition is direct and
 567 tighter — and the support condition $Q_t \geq q_0$ is *not* required for this term, only for conclusion (iii).)
 568 Vanishes in Regime A.

569 Combining Steps 1–4 gives conclusion (v): $\mathbb{E}[\text{DynReg}(T)] \leq 2cV_{\max}N_h \sqrt{(B_T + 1)T} + V_{\max}N_h(1 -$
 570 $p_{\text{id}}) \cdot T$. The Cesàro tracking corollary $\frac{1}{T} \sum_t (V_t^{\tilde{\pi}_t^*} - V_t^{Q_t}) = O(V_{\max}N_h \sqrt{(B_T + 1)/T}) \rightarrow 0$ (when
 571 $B_T = o(T)$ and $p_{\text{id}} \rightarrow 1$, i.e., Regime A) follows by dividing through by T .

572 **Bandit case sharpening** In the bandit special case ($N_h = 1$), Thompson sampling and UCB
 573 satisfy (A5) with the *stronger logarithmic rate* $\mathbb{E}[1 - e^{-K_t}] = O(\log t / (t \Delta_{\min}^2))$ per stationary block
 574 (Appendix D) — by Lattimore and Szepesvári 2020^[27] Theorem 7.1, $\mathbb{E}[N_a(L)] = O(\log L / \Delta_a^2)$ per
 575 suboptimal arm, aggregating to the per-round form. Cumulative regret $O(V_{\max}(B_T + 1) \log(T / (B_T +$
 576 $1)) / \Delta_{\min}^2)$ — sharper than the worst-case $\sqrt{(B_T + 1)T}$ when $\Delta_{\min}$ is bounded away from zero. The
 577 framework’s $V_{\max} \cdot \text{TV}$ chain is structurally one factor of $\Delta_{\min}$ looser than direct gap-aware UCB
 578 analysis (which gives $O(\log T / \Delta_{\min})$ cumulative by weighting each suboptimal pull by its own $\Delta(a)$
 579 rather than by $V_{\max}$); this is a feature of the unification — the framework’s contribution is composition,
 580 not a sharper bandit constant.

581 **Carryover variant of (A5)** If the base learner does *not* restart at block boundaries — e.g., a single
 582 UCRL2 instance run across the whole horizon — the within-block $\sqrt{L}$ guarantee can fail at switch
 583 events because $d_h^{Q_t}$ at the start of block $i + 1$ reflects the previous block’s policy rather than a fresh
 584 exploration distribution. Bounding the cross-block occupancy shift requires either (a) detecting

switches and locally re-exploring (which recovers a restart-style analysis), or (b) showing the base learner’s regret is robust to mis-calibrated state distributions. Sliding-window or adaptive-window variants^[1,2] can be analyzed this way under additional structural assumptions. Theorem 4.1(v) is stated for the restart case; the carryover case is plausible but requires bounding the occupancy shift, deferred as future work.

Detailed full proof of (v) in Appendix A.

6 Conclusion

We assemble four components — the two-gap diagnostic, the point-mass reverse-KL/TV identity, the multi-factor strategic-tempo forgetting prerequisite, and closed-loop interventional access — into a unified non-stationary RL convergence theory. The composition delivers three properties no published strand has individually: it *handles non-stationarity* (via the strategic-tempo persistence inequality), has *explicit metric structure on policy space* (via the point-mass identity), and is *learnable on-policy* (via closed-loop interventional access). The cumulative dynamic regret rate $\tilde{O}(N_h \sqrt{(B_T + 1)T})$ matches the piecewise-stationary B_T family of^[1] as a corollary; the per-round identity coordinate is *exact* under deterministic π^* and strictly sharper than Pinsker and Bretagnolle–Huber at this corner. The point-mass identity is the technical anchor — absent, to our knowledge, from the prior RL/non-stationary corpus as an upper-bound regret coordinate.

We close with concluding observations and practitioner takeaways.

6.1 Concluding observations

On the optimal dependencies on N_h , S , A The lifted UCRL2/UCBVI rate $\tilde{O}(N_h^2 \sqrt{SA(B_T + 1)T})$ has linear-in- N_h per-round penalty (the simulation lemma) plus an additional N_h from the within-block constant c . Bernstein-type concentration in the base learner could potentially shave one $\sqrt{N_h}$ factor, by analogy with^[43] for tabular non-stationary MDPs. We believe the $\sqrt{N_h}$ gap between this and the corresponding tabular non-stationary lower bound is expected because the base-learner concentration used here is intrinsically Hoeffding-type.

On the continuous-variation extension The current rate is in the piecewise-stationary B_T family (kernel-stationary-segment count, Section 3). The same per-round identity route extends to the continuous-variation V_T regime via the black-box reduction of^[2] (MASTER), which wraps any stationary base learner with $\tilde{O}(\sqrt{L})$ regret into a non-stationary algorithm achieving $\tilde{O}(\min\{\sqrt{(B_T + 1)T}, (V_T + 1)^{1/3}T^{2/3}\})$ Best-of-Both-Worlds dynamic regret without prior knowledge of either variation budget. Our framework’s identity-routed (A5)-compatible learner — achieving $\sum_{t \in \text{block}} \mathbb{E}[\text{TV}_t] \leq 2c\sqrt{L}$ in TV coordinates per stationary block — satisfies MASTER’s base-learner condition (Wei-Luo Assumption 1, with $C(t) = ct^{1/2}$, $p = 1/2$ at the boundary case explicitly admitted). The wrapping is mechanical: the per-round identity (Steps 1–2 of Appendix A) is stationarity-agnostic and survives the substitution of block-Cauchy–Schwarz aggregation by adaptive-window MASTER aggregation; the misidentification penalty (Step 4) is round-additive and rides through. Theorem-grade availability in Appendix A (Remark on (A5’)-BoBW). The achieved $V_T^{1/3}T^{2/3}$ exponent matches^[3]’s near-optimal RestartQ-UCB rate $\tilde{O}((SA)^{1/3}\Delta^{1/3}HT^{2/3})$ and is information-theoretically near-optimal: the^[47] lower bound $\Omega(K^{1/3}V_T^{1/3}T^{2/3})$ for non-stationary stochastic bandits applies *also at the deterministic- π corner** (their hard instance is a Bernoulli MAB with dynamic oracle playing the per-round argmax — deterministic-per-round optimum), so the per-round identity sharpens *constants and per-round form*, not the V_T exponent. The two regimes are *not* structurally distinct: they share the same per-round coordinate $1 - e^{-K_t}$ and differ only in *aggregation strategy* (block-Cauchy–Schwarz across $B_T + 1$ stationary intervals vs. adaptive-window MASTER over $\log T$ parallel base-learner instances). MASTER also gives the parameter-free property — automatic adaptation to whichever budget is sharper without prior knowledge of B_T or V_T — that Mao 2021’s basic RestartQ-UCB requires^[3]’s Double-Restart variant for. Two refinements remain open: an *agent-experienced* continuous-variation budget $V_T^{\text{eff}, \Sigma}$ paralleling the $B_T^{\text{eff}, \Sigma}$ direction below, and an *identity-coordinate* variation budget $V_T^{(K)} := \sum_t |K_t - K_{t-1}|$ that lives on the action-distribution

coordinate and could give sharper-than- V_T rates in absorbing regimes where the strategic substate damps environment-side drift.

On agent-experienced block count The B_T of the rate (v) is an *environment-side* quantity — every kernel-and-reward change counts. The strategic-tempo guarantee (ii) suggests an *agent-experienced* block count $B_T^{\text{eff}, \Sigma}$ — the number of times $\|\delta_\Sigma\|$ exits its ultimate-bounded region — could be strictly smaller than B_T in regimes where per-element forgetting absorbs within-segment drift. Whether (v) sharpens to $\sqrt{(B_T^{\text{eff}, \Sigma} + 1) T}$ requires algorithmic design coupling the base learner’s confidence-set widening with the strategic substate δ_Σ , deferred to follow-up. The result would explain why agents in high- V_T slow-drift environments often outperform B_T -pessimistic predictions: what matters is the *agent’s effective* variation budget, not the environment’s.

On joint uniqueness of the four-component composition The metric layer is uniquely reverse-KL up to positive scaling under chain-rule additivity (Theorem C.3 in Appendix C.10) — any theory satisfying explicit metric structure plus chain-rule must use reverse-KL. We do *not* claim joint uniqueness across all three properties (non-stationarity / metric / learnable). Alternative correction architectures (non-sector-bounded revisions; non-diagonal correction) and alternative interventional-access taxonomies exist in principle, and a joint-uniqueness theorem would require further axioms (singular-trajectory + sector-boundedness + identifiability conditions) — open.

On coupled-goal architectures (C2)’s sequential-ignorability — that H_t d-separates a_t from o_{t+1} in the mutilated graph $G_{\bar{a}_t}$ — presupposes architectural separation (conditional independence) between the model state M_t (in H_t) and the goal state. Goal-conditioned LLM policies and similar coupled-goal architectures violate this separation: the goal influences both action selection (via the policy) and the environment’s response to the action (via shared latent context such as user intent, dialogue history, or task framing), so H_t does not block the goal-mediated path from a_t to o_{t+1} — (C2) fails and the loop’s data-generating process is no longer interventional in our sense. The conditional-independence relaxation results of^[48] are a starting point for handling such cases. We treat this as an interesting boundary case that the current framework does not directly cover.

On strict tightness of the strategic-tempo threshold The bottleneck threshold $\mathcal{T}_\Sigma^{\text{bn}, \text{ss}} > \rho_\Sigma / R_\Sigma$ is *sufficient and sharp inside the diagonal sector model* — when reversed, an adversarial disturbance concentrating on the bottleneck element drives the modeled mismatch beyond R_Σ . The theorem is silent about correction architectures *outside* the sector-Lyapunov reduction (e.g., non-diagonal correction, alternative non-sector-bounded revisions, or stabilization produced by mechanisms other than the modeled α_Σ -correction). Whether the threshold extends to those architectures is a useful open question in the adaptive-control-RL bridge.

6.2 Practitioner takeaways

For a practitioner seeking to apply the four-component theory:

1. **The point-mass identity at the deterministic- π^* corner is a corner-case sharpening of Bretagnolle–Huber that hasn’t been deployed in RL regret upper bounds.** Where deterministic π^* applies — most tabular finite-MDP regret analyses with isolated optima — the identity yields *exact* regret-vs-KL Lipschitz equivalence rather than the loose Pinsker / BH inequalities. The per-round coordinate $1 - e^{-D_{\text{KL}}}$ is computable directly from Q and is strictly sharper than $\sqrt{D_{\text{KL}}/2}$ at every $D_{\text{KL}} > 0$.
2. **The multi-factor strategic tempo lifts ProST’s tempo-as-hyperparameter framing into a structural survival inequality.** Three factors (observation rate ν , identifiability ι , discount rate $1 - \lambda$) are independently load-bearing per element. A practitioner can use the aggregate necessary-condition $\mathcal{T}_\Sigma^{\text{agg}, \text{ss}} > |E| \cdot \rho_\Sigma / R_\Sigma$ as a fail-fast pre-check and the variation-budget estimation literature^[1,2,39] to estimate the environment-side ρ_Σ / R_Σ from observed reward and transition variation.
3. **Closed-loop interventional access makes regret bounds learnable on-policy under sufficient identifiability.** Regime A (intervention-rich, $\iota \approx 1$) makes the bound directly realizable; Regime B (partial) carries quantified bias linear in $1 - p_{\text{id}}$; Regime C (observation-only)

requires observer-on-sub-agent extensions (Appendix C.8) or flags the regime as out-of-scope. The 2×2 diagnostic routes corrective action when regret is non-zero — distinguishing “policy revision” from “model / policy class / horizon revision” from “objective revision.”

We hope this work contributes a unifying lens for the four-track non-stationary RL landscape, and that the point-mass identity is a useful tool in subsequent regret analyses where deterministic π^* is the canonical scope.

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801 A Proof of the Composition Theorem

802 This appendix gives the full proof of Theorem 4.1. The proof composes the four key lemmas of
 803 Section 5 (full proofs of those in Appendix B) with a block decomposition at optimum-change events
 804 plus Cauchy–Schwarz across blocks.

805 **Notation** Throughout this section: $K_t(s) := D_{\text{KL}}(\delta_{a_t^*(s)} \parallel Q_t(\cdot \mid s))$ is the per-state reverse-KL
 806 coordinate; $d_h^{Q_t}(\cdot \mid s_0)$ is the round- t learner-induced state distribution at horizon step h starting from
 807 s_0 ; $\overline{\text{TV}}_t := \frac{1}{N_h} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}} [\text{TV}(\pi_t^*(\cdot \mid s_h), Q_t(\cdot \mid s_h))]$ is the occupancy-weighted per-round
 808 coordinate of (A5); $B_T = |\{t : (P_t, r_t) \neq (P_{t-1}, r_{t-1})\}|$ counts piecewise-stationary segment
 809 boundaries (per Section 3); the partition is $0 = \tau_0 < \tau_1 < \dots < \tau_{B_T} \leq \tau_{B_T+1} = T$ with block i
 810 length $\Delta_i := \tau_{i+1} - \tau_i$.

811 A.1 Proof of (i) — Per-round identity-bounded regret

812 For deterministic $\pi_t^* = \delta_{a_t^*(s)}$ and any $Q_t(\cdot | s)$, Key Lemma 1 (Lemma 5.1, full proof in Ap-
 813 pendix B.1) gives the exact identity $\text{TV}(\delta_{a_t^*(s)}, Q_t(\cdot | s)) = 1 - e^{-K_t(s)}$ in the extended real sense
 814 (when $Q_t(a_t^*(s) | s) = 0$, $K_t(s) = +\infty$ and both sides equal 1, recovering the trivial bound).
 815 Composed with the per-state TV-regret bounds (full proofs in Appendix B.1 and Appendix C.4):
 816 $\Delta_{\min} \cdot \text{TV}(\delta_{a_t^*(s)}, Q_t(\cdot | s)) \leq R(Q_t | s) \leq V_{\max} \cdot \text{TV}(\delta_{a_t^*(s)}, Q_t(\cdot | s))$, substituting the identity
 817 yields conclusion (i): $\Delta_{\min}(1 - e^{-K_t(s)}) \leq R(Q_t | s) \leq V_{\max}(1 - e^{-K_t(s)})$. $\square$

818 A.2 Proof of (ii) — Aggregate mismatch ultimate boundedness

819 Under (A2), Key Lemma 2 (Lemma 5.2, full proof in Appendix B.3) applied to the diagonal sector
 820 model with $\alpha_{ij}^{\text{ss}} = \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij})$ gives ultimate boundedness of $\|\delta_{\Sigma}\|$ within $R_{\Sigma}^* = \rho_{\Sigma} / \mathcal{T}_{\Sigma}^{\text{bn,ss}}$. $\square$

821 A.3 Proof of (iii) — KL coordinate bias bound

822 Key Lemma 4 (Lemma 5.4, full proof in Appendix B.6) gives the per-state bias bound directly, under
 823 (A3)’s identifiability assumption (which gives $p_{\text{id}}(s)$ for each s) and the two-point support condition
 824 $Q_t(\cdot | s) \geq q_0$ at $a_{\text{ag},t}^*(s)$ and $\tilde{a}_t^*(s)$. $\square$

825 A.4 Proof of (iv) — 2×2 corrective-action routing

826 (A4) states the agent applies the satisfaction-gap / control-regret 2×2 (Section 4.1 Component 1).
 827 Conclusion (iv) is a direct restatement of the 2×2 cell partition. The decomposition is convention-
 828 dependent (the values of δ_{sat} and δ_{regret} depend on the continuation convention; see Appendix C.1 for
 829 monotonicity across C1/C2/C3); the 2×2 structure is preserved across all three conventions. $\square$

830 A.5 Proof of (v) — Trajectory-level cumulative regret

831 We chain Key Lemmas 1 and 4 with the simulation lemma and a Cauchy–Schwarz block argument.
 832 The proof is in four steps; the high-level sketch is in Section 5.5.

833 *Step 1: Simulation / performance-difference lemma.* For any two policies π
 834 and π' , any state s , and finite-horizon non-discounted MDP with horizon N_h ,
 835 the standard performance-difference lemma^[40–43] gives $V_t^{\pi'}(s) - V_t^{\pi}(s) =$
 836 $\mathbb{E}\left[\sum_{h=0}^{N_h-1} \left(\mathbb{E}_{a' \sim \pi'(\cdot | s_h)}[Q_h^{\pi'}(s_h, a')] - Q_h^{\pi}(s_h, a_h)\right) \mid s_0 = s, a_h \sim \pi\right]$. Each per-step bracket
 837 is bounded by value range times TV: $|\mathbb{E}_{a' \sim \pi'}[Q_h^{\pi'}] - \mathbb{E}_{a \sim \pi}[Q_h^{\pi}]| \leq V_{\max} \cdot \text{TV}(\pi'(\cdot |$
 838 $s_h), \pi(\cdot | s_h))$. Applied at round t with $\pi' = \pi_t^*$, $\pi = Q_t$: $V_t^{\pi_t^*}(s_0) - V_t^{Q_t}(s_0) \leq$
 839 $V_{\max} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}(\cdot | s_0)}[\text{TV}(\pi_t^*(\cdot | s_h), Q_t(\cdot | s_h))] = V_{\max} \cdot N_h \cdot \overline{\text{TV}}_t$. The horizon factor N_h
 840 here is the *linear-in- N_h* simulation-lemma penalty — sharper than the N_h^2 of TRPO-style worst-state
 841 bounds^[44], which bound max-state TV rather than occupancy-weighted TV.

842 *Step 2: Per-state TV identity along the trajectory.* By the deterministic-per-state optimum scope of
 843 Theorem 4.1 ($\pi_t^*(\cdot | s) = \delta_{a_t^*(s)}$), at every visited state Key Lemma 1 gives $\text{TV}(\pi_t^*(\cdot | s), Q_t(\cdot |$
 844 $s)) = 1 - Q_t(a_t^*(s) | s) = 1 - e^{-K_t(s)}$. Summing over the trajectory and taking expectations:
 845 $\overline{\text{TV}}_t = \frac{1}{N_h} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}}[1 - e^{-K_t(s_h)}]$. The per-state KL/TV identity is preserved underneath
 846 the occupancy aggregation; the per-round coordinate is sharper than Pinsker/BH at every visited state.

847 *Step 3: Block-sum + Cauchy–Schwarz across blocks.* Within each block $[\tau_i, \tau_{i+1})$ of length
 848 Δ_i , (A5)’s restart-on-change identity-tight base learner gives $\sum_{t \in \text{block}_i} \mathbb{E}[\overline{\text{TV}}_t] \leq 2c\sqrt{\Delta_i}$. Sum-
 849 ming over the $B_T + 1$ blocks: $\sum_{t=1}^T \mathbb{E}[\overline{\text{TV}}_t] = \sum_{i=0}^{B_T} \sum_{t \in \text{block}_i} \mathbb{E}[\overline{\text{TV}}_t] \leq 2c \sum_{i=0}^{B_T} \sqrt{\Delta_i}$.
 850 Cauchy–Schwarz gives $\sum_{i=0}^{B_T} \sqrt{\Delta_i} \leq \sqrt{(B_T + 1) \cdot \sum_i \Delta_i} = \sqrt{(B_T + 1)T}$. Multiplying by
 851 $V_{\max} N_h$: $\sum_{t=1}^T \mathbb{E}[V_{\max} N_h \overline{\text{TV}}_t] \leq 2c V_{\max} N_h \sqrt{(B_T + 1)T}$. This is the rate term.

852 *Step 4: Misidentification penalty via simulation lemma at the identified-vs-true-optimum gap.* Steps
 853 1–3 implicitly compared the agent’s policy Q_t to the agent’s identified-optimum policy $\pi_{\text{ag},t}^*(\cdot |$

854 $s) := \delta_{a_{\text{ag},t}^*(s)}$ (since (A5)’s base-learner guarantee is what the agent achieves on its identified
855 objective). The trajectory-level dynamic regret in (v) compares to the *true* optimum $\tilde{\pi}_t^*(\cdot | s) := \delta_{\tilde{a}_t^*(s)}$.
856 Decompose: $V_t^{\tilde{\pi}_t^*}(s_0) - V_t^{Q_t}(s_0) = [V_t^{\tilde{\pi}_t^*}(s_0) - V_t^{\pi_{\text{ag},t}^*}(s_0)] + [V_t^{\pi_{\text{ag},t}^*}(s_0) - V_t^{Q_t}(s_0)]$. Steps
857 1–3 bound the second bracket. The first bracket is the misidentification penalty. Apply the simulation
858 lemma to the two point-mass policies $(\tilde{\pi}_t^*, \pi_{\text{ag},t}^*)$: each per-step bracket reduces to $Q_h^{\tilde{\pi}_t^*}(s_h, \tilde{a}_t^*(s_h)) -$
859 $Q_h^{\pi_{\text{ag},t}^*}(s_h, a_{\text{ag},t}^*(s_h)) \in [0, V_{\max}] \cdot \mathbf{1}[\tilde{a}_t^*(s_h) \neq a_{\text{ag},t}^*(s_h)]$. Lower bound by optimality of $\tilde{a}^*$; upper
860 bound by the cumulative-Q range with the indicator on the misidentification event. Taking expectations,
861 with the per-state floor $p_{\text{id}} := \min_s p_{\text{id}}(s)$: $\mathbb{E}[V_t^{\tilde{\pi}_t^*}(s_0) - V_t^{\pi_{\text{ag},t}^*}(s_0)] \leq V_{\max} \cdot N_h \cdot (1 - p_{\text{id}})$. Summing
862 over T rounds: $V_{\max} \cdot N_h \cdot (1 - p_{\text{id}}) \cdot T$. This is the bias term, in *value coordinates* throughout — no
863 detour through the KL coordinate. Key Lemma 4’s KL-readout bound is the right tool for conclusion
864 (iii)’s diagnostic computability; for the value-coordinate misidentification penalty here, the indicator
865 decomposition is direct and tighter (the support condition $Q_t \geq q_0$ at the two argmax candidates is
866 *not* required for this term — only conclusion (iii) needs it).

867 *Combining.* Adding Steps 1–4: $\mathbb{E}[\text{DynReg}(T)] \leq 2c V_{\max} N_h \sqrt{(B_T + 1)T} + V_{\max} N_h (1 - p_{\text{id}}) T$,
868 which is conclusion (v). $\square$

869 **Cesàro tracking corollary** Dividing through by T : $\frac{1}{T} \sum_t (V_t^{\tilde{\pi}_t^*} - V_t^{Q_t}) \leq$
870 $2c V_{\max} N_h \sqrt{(B_T + 1)/T} + V_{\max} N_h (1 - p_{\text{id}})$. As $T \rightarrow \infty$ with $B_T = o(T)$ and $p_{\text{id}} \rightarrow 1$
871 (Regime A), the right side tends to zero — the agent’s average value over the trajectory tracks the
872 moving optimum’s average value. In Regime B the second term is constant; tracking is only up to a
873 $V_{\max}(1 - p_{\text{id}})$ residual. $\square$

874 **Bandit case sharpening (Remark)** In the bandit special case ($N_h = 1$, so $\overline{\text{TV}}_t = \mathbb{E}[1 - e^{-K_t}]$),
875 Thompson sampling and UCB (Appendix D) satisfy (A5) with the *stronger logarithmic* per-block rate.
876 By Lattimore and Szepesvári 2020^[27] Theorem 7.1 (or [? auer-cesa-bianchi-fischer-2002-finitetime]),
877 $\mathbb{E}[N_a(L)] = O(\log L / \Delta_a^2)$ per suboptimal arm a ; aggregating over arms gives $\mathbb{E}[1 - e^{-K_t}] =$
878 $\mathbb{E}[1 - Q_t(a^*)] = O(\log t / (t \Delta_{\min}^2))$ per stationary block. Substituting and summing gives cumulative
879 regret $O(V_{\max}(B_T + 1) \log(T/(B_T + 1)) / \Delta_{\min}^2)$ — sharper than the worst-case $\sqrt{(B_T + 1)T}$ when
880 $\Delta_{\min}$ is bounded away from zero. The framework’s $V_{\max} \cdot \text{TV}$ chain is structurally one factor of $\Delta_{\min}$
881 looser than direct gap-aware UCB analysis (which gives $O(\log T / \Delta_{\min})$ cumulative regret per block
882 by weighting each suboptimal pull by $\Delta(a)$ rather than $V_{\max}$); this is a feature of the unification, not a
883 bug. For $N_h > 1$ MDPs, UCRL2 and UCBVI provide the natural occupancy-weighted instantiation
884 (Appendix D) with c of order $\tilde{O}(\sqrt{N_h SA})$, lifting cleanly to $\tilde{O}(N_h^2 \sqrt{SA(B_T + 1)T})$ — same
885 piecewise-stationary B_T family as restart-on-change variants of^[1].

886 **Best-of-Both-Worlds non-stationarity extension (Remark on (A5’))** Conclusion (v) above uses
887 the restart-on-change form of (A5). The per-round identity route is *stationarity-agnostic* — Steps 1,
888 2, 4 above hold per round regardless of cross-round dynamics; only Step 3’s block-Cauchy–Schwarz
889 aggregator is piecewise-stationary-specific. Replacing (A5) by

890 **(A5’) MASTER-wrapped base learner.** The base learner is the Wei and Luo 2021^[2]
891 MASTER black-box wrapping of any base algorithm satisfying their Assumption 1
892 (a near-stationary regret guarantee with $C(t) = c_1 t^p + c_2$ for $p \in [1/2, 1)$, plus an
893 auxiliary-quantity output condition; UCRL2, UCBVI, and Thompson sampling all
894 qualify), giving stationary regret $\sum_{t \in W} \mathbb{E}[\text{TV}_t] \leq 2c\sqrt{|W|}$ within each MASTER-
895 selected window W .

896 substitutes Step 3’s block-Cauchy–Schwarz aggregator with MASTER’s adaptive-window aggrega-
897 tor (parallel base-learner instances at exponentially-spaced window sizes $W_k = 2^k$, online meta-
898 learner over the $\log T$ instances). The wrapping is mechanical: the per-round identity (Steps 1–2)
899 is preserved unchanged at each round; the misidentification penalty (Step 4) is round-additive and
900 rides through; only the aggregator changes. The resulting rate, by Wei-Luo Theorem 2 applied
901 with our framework’s $C(t) = 2c V_{\max} N_h \sqrt{t}$ at the boundary $p = 1/2$, is $\mathbb{E}[\text{DynReg}(T)] \leq$
902 $\tilde{O}\left(V_{\max} N_h \cdot \min \left\{ \sqrt{(B_T + 1)T}, (V_T + 1)^{1/3} T^{2/3} \right\} \right) + V_{\max} N_h (1 - p_{\text{id}}) T$, automatically —

without prior knowledge of B_T or V_T . The $V_T^{1/3}T^{2/3}$ exponent matches^[3]’s near-optimal RestartQ-UCB rate and is information-theoretically near-optimal: the^[47] lower bound $\Omega(K^{1/3}V_T^{1/3}T^{2/3})$ for non-stationary stochastic bandits applies *also at the deterministic- π corner** (their construction is a Bernoulli MAB with dynamic oracle playing the per-round argmax — deterministic-per-round optimum, identical to our canonical scope), so the per-round identity sharpens *constants and per-round form*, not the V_T exponent. The two regimes are *not* structurally distinct: they share the per-round coordinate $1 - e^{-K_t}$ and differ only in the aggregation strategy. $\square$

910 B Proofs of the Key Lemmas

911 This appendix gives full proofs of the four key lemmas surfaced in Section 5, plus the perturbative extension and the ProST sector-level reduction.

913 B.1 Proof of Key Lemma 1 (Point-mass reverse-KL/TV identity)

914 For deterministic $\pi^* = \delta_{a^*}$ and any policy Q (extending $D_{\text{KL}} = +\infty$ when $Q(a^*) = 0$, which gives
915 $1 - e^{-D_{\text{KL}}} = 1 = \text{TV}$ trivially):

916 The reverse-KL collapses under the point-mass: $D_{\text{KL}}(\delta_{a^*} \| Q) = \sum_a \delta_{a^*}(a) \log \frac{\delta_{a^*}(a)}{Q(a)} =$
917 $\log \frac{1}{Q(a^*)} = -\log Q(a^*)$, where the convention $0 \log 0 = 0$ disposes of the $a \neq a^*$ terms.
918 Combined with the textbook total-variation calculation $\text{TV}(\delta_{a^*}, Q) = \frac{1}{2} \sum_a |\delta_{a^*}(a) - Q(a)| =$
919 $\frac{1}{2} (|1 - Q(a^*)| + \sum_{a \neq a^*} Q(a)) = \frac{1}{2} ((1 - Q(a^*)) + (1 - Q(a^*))) = 1 - Q(a^*)$, we have
920 $-\log Q(a^*) = -\log(1 - \text{TV})$. Solving for TV gives $\text{TV} = 1 - e^{-D_{\text{KL}}}$. $\square$

921 **Strict-improvement substitution** Substituting the identity into the Bretagnolle–Huber inequality
922 $\text{TV}(P, Q) \leq \sqrt{1 - e^{-D_{\text{KL}}(P \| Q)}}$ at $P = \delta_{a^*}$ yields $1 - e^{-D_{\text{KL}}} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$, which holds *strictly*
923 on $D_{\text{KL}} > 0$ since $x < \sqrt{x}$ on $(0, 1)$. The identity therefore supplies the *exact* TV value at the
924 deterministic- π^* corner and lies strictly below the BH envelope there.

925 **Two-sided regret bound** (stated as Component 2’s two-sided characterization in Section 4.1 and as Key
926 Lemma 1’s *Per-round regret bound* paragraph in Section 5.1). From the regret expression $R(Q) =$
927 $\sum_{a \neq a^*} Q(a) \Delta(a)$ and $\Delta(a) \leq V_{\max}$: $R(Q) \leq V_{\max} \sum_{a \neq a^*} Q(a) = V_{\max} (1 - Q(a^*)) =$
928 $V_{\max} \text{TV}(\delta_{a^*}, Q) = V_{\max} (1 - e^{-D_{\text{KL}}})$. The matching action-gap lower bound (full statement in
929 Appendix C.4) gives $R(Q) = \sum_{a \neq a^*} Q(a) \Delta(a) \geq \Delta_{\min} \sum_{a \neq a^*} Q(a) = \Delta_{\min} \text{TV}(\delta_{a^*}, Q) =$
930 $\Delta_{\min} (1 - e^{-D_{\text{KL}}})$. Combining gives the two-sided characterization. The bound is *Lipschitz-equivalent*
931 with constants $\Delta_{\min}/V_{\max}$ and 1, both achieved on extremal value landscapes — therefore *coordinate-*
932 *optimal among bounds depending only on TV*. $\square$

933 B.2 Proof of the perturbative extension — ϵ -stochastic and softmax-regularized

934 We establish the perturbative identity via a uniform perturbation argument rather than an alignment-
935 specific calculation.

936 **ϵ -greedy stochastic optimum** Let $\pi_\epsilon^*(a^*) = 1 - \epsilon$ and $\pi_\epsilon^*(a) = \epsilon/(|\mathcal{A}| - 1)$ for $a \neq a^*$, and
937 assume the full-support lower bound $Q(a) \geq q_0 > 0$ for all $a \in \mathcal{A}$. (Full support is required since π_ϵ^*
938 has positive mass on every action; without $Q(a) > 0$ on the full support, $D_{\text{KL}}(\pi_\epsilon^* \| Q) = +\infty$.)

939 *Step 1 — TV is Lipschitz under the point-mass perturbation.* $\text{TV}(\pi_\epsilon^*, \delta_{a^*}) = \epsilon$. By the triangle
940 inequality for total variation, $|\text{TV}(\pi_\epsilon^*, Q) - \text{TV}(\delta_{a^*}, Q)| \leq \text{TV}(\pi_\epsilon^*, \delta_{a^*}) = \epsilon$, uniformly in Q —
941 no alignment hypothesis needed.

942 *Step 2 — Reverse-KL admits a uniform expansion.* Direct computation:

$$\begin{aligned} D_{\text{KL}}(\pi_\epsilon^* \| Q) - D_{\text{KL}}(\delta_{a^*} \| Q) &= (1 - \epsilon) \log \frac{1 - \epsilon}{Q(a^*)} + \frac{\epsilon}{|\mathcal{A}| - 1} \sum_{a \neq a^*} \log \frac{\epsilon/(|\mathcal{A}| - 1)}{Q(a)} - (-\log Q(a^*)) \\ &= (1 - \epsilon) \log(1 - \epsilon) + \epsilon \log Q(a^*) + \epsilon \log \frac{\epsilon}{|\mathcal{A}| - 1} - \frac{\epsilon}{|\mathcal{A}| - 1} \sum_{a \neq a^*} \log Q(a), \end{aligned}$$

where the second equality collects the $\log Q(a^*)$ contributions via $-(1-\epsilon)\log Q(a^*) + \log Q(a^*) = \epsilon\log Q(a^*)$ and splits the constant $\log$ term off the conditional sum. Expanding $(1-\epsilon)\log(1-\epsilon) = -\epsilon + O(\epsilon^2)$ and $\epsilon\log(\epsilon/(|\mathcal{A}| - 1)) = \epsilon\log\epsilon - \epsilon\log(|\mathcal{A}| - 1)$, the leading-order behavior is $D_{\text{KL}}(\pi_\epsilon^* \| Q) = D_{\text{KL}}(\delta_{a^*} \| Q) + \epsilon\log\epsilon - \epsilon(\log(|\mathcal{A}| - 1) + 1) + \epsilon\log Q(a^*) - \frac{\epsilon}{|\mathcal{A}|-1} \sum_{a \neq a^*} \log Q(a) + O(\epsilon^2)$. Under $Q(a) \geq q_0$, each $|\log Q(a)| \leq \log(1/q_0)$ is uniformly bounded; the dominant correction is the $\epsilon\log\epsilon$ leading term, giving $|D_{\text{KL}}(\pi_\epsilon^* \| Q) - D_{\text{KL}}(\delta_{a^*} \| Q)| = O(\epsilon\log(1/\epsilon) + \epsilon\log(1/q_0)) = O(\epsilon\log(1/\epsilon))$ absorbing $\epsilon\log(1/q_0)$ into the leading $\epsilon\log(1/\epsilon)$ term as $\epsilon \rightarrow 0$ (constants depend on q_0 and $|\mathcal{A}|$).

Step 3 — Map through $1 - e^{-x}$ via Lipschitzness. The map $x \mapsto 1 - e^{-x}$ is 1-Lipschitz on $[0, \infty)$, so $|(1 - e^{-D_{\text{KL}}(\pi_\epsilon^* \| Q)}) - (1 - e^{-D_{\text{KL}}(\delta_{a^*} \| Q)})| = O(\epsilon\log(1/\epsilon))$. Combining Steps 1 and 3 with the unperturbed identity $\text{TV}(\delta_{a^*}, Q) = 1 - e^{-D_{\text{KL}}(\delta_{a^*} \| Q)}$: $\text{TV}(\pi_\epsilon^*, Q) = 1 - e^{-D_{\text{KL}}(\pi_\epsilon^* \| Q)} + O(\epsilon\log(1/\epsilon))$ uniformly in Q over the class $\{Q : Q(a) \geq q_0\}$, with constants depending on q_0 and $|\mathcal{A}|$. This establishes the ϵ -greedy form of the perturbative identity. $\square$

Softmax-regularized optimum Let $\pi_\tau^*(a) \propto \exp(Q_O(a)/\tau)$. Under isolated optimum with action gap $\Delta_{\min}$, the off-optimum mass concentrates exponentially in $1/\tau$: $1 - \pi_\tau^*(a^*) = \frac{\sum_{a \neq a^*} \exp(-(Q_O(a^*) - Q_O(a))/\tau)}{1 + \sum_{a \neq a^*} \exp(-(Q_O(a^*) - Q_O(a))/\tau)} \leq (|\mathcal{A}| - 1) \exp(-\Delta_{\min}/\tau)$. Treating π_τ^* as π_ϵ^* with effective $\epsilon = O(\exp(-\Delta_{\min}/\tau))$, the ϵ -greedy expansion above applies with $\epsilon\log(1/\epsilon) = O((\Delta_{\min}/\tau) \exp(-\Delta_{\min}/\tau)) = O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$, exponentially small with a polynomial prefactor (equivalently, $O(\exp(-c\Delta_{\min}/\tau))$ for any fixed $c \in (0, 1)$ since the exponential dominates the polynomial). $\square$

Two-sided regret bound under perturbation Composing with the TV-regret bounds: $\Delta_{\min}(1 - e^{-D_{\text{KL}}(\pi_\epsilon^* \| Q)}) - O(\epsilon\log(1/\epsilon)) \leq R(Q) \leq V_{\max}(1 - e^{-D_{\text{KL}}(\pi_\epsilon^* \| Q)}) + O(\epsilon\log(1/\epsilon))$.

B.3 Proof of Key Lemma 2 (Multi-factor forgetting prerequisite)

We work in the continuous-time idealization of Model (Σ): $\dot{\delta}_\Sigma = -\mathbf{F}(\delta_\Sigma) + \mathbf{w}$ with $\mathbf{F}(\delta_\Sigma) := (\alpha_{ij}^{\text{ss}} \delta_{ij})_{(i,j) \in E}$ the sector correction and $\mathbf{w}$ the deterministic disturbance with $\|\mathbf{w}\| \leq \rho_\Sigma$. The discrete-time form recovers the same scaling up to absorbable constants via an AM-GM expansion of $\|\delta_\Sigma - \mathbf{F} + \mathbf{w}\|^2$ (sketched below).

For the Lyapunov function $V(\delta_\Sigma) := \|\delta_\Sigma\|^2$, $\dot{V} = 2\delta_\Sigma^\top \dot{\delta}_\Sigma = -2\delta_\Sigma^\top \mathbf{F}(\delta_\Sigma) + 2\delta_\Sigma^\top \mathbf{w}$. The first term is the *sector product*, bounded below by the bottleneck: $\delta_\Sigma^\top \mathbf{F}(\delta_\Sigma) = \sum_{(i,j)} \alpha_{ij}^{\text{ss}} \delta_{ij}^2 \geq \min_{(i,j)} \alpha_{ij}^{\text{ss}} \cdot \|\delta_\Sigma\|^2 = \mathcal{T}_\Sigma^{\text{bn,ss}} \cdot \|\delta_\Sigma\|^2$, with the bound tight: setting δ_Σ proportional to $\mathbf{e}_{(i^*, j^*)}$ where $(i^*, j^*) \in \arg \min_{(i,j)} \alpha_{ij}^{\text{ss}}$ saturates it. The cross-term is bounded by Cauchy-Schwarz: $|2\delta_\Sigma^\top \mathbf{w}| \leq 2\|\delta_\Sigma\| \cdot \|\mathbf{w}\| \leq 2\rho_\Sigma \|\delta_\Sigma\|$. Combining: $\dot{V} \leq -2\mathcal{T}_\Sigma^{\text{bn,ss}} \|\delta_\Sigma\|^2 + 2\rho_\Sigma \|\delta_\Sigma\| = -2\|\delta_\Sigma\| \cdot (\mathcal{T}_\Sigma^{\text{bn,ss}} \|\delta_\Sigma\| - \rho_\Sigma)$. The drift is strictly negative whenever $\|\delta_\Sigma\| > \rho_\Sigma / \mathcal{T}_\Sigma^{\text{bn,ss}}$; the trajectory is ultimately bounded by $R_\Sigma^* = \rho_\Sigma / \mathcal{T}_\Sigma^{\text{bn,ss}}$. So whenever $\mathcal{T}_\Sigma^{\text{bn,ss}} > \rho_\Sigma / R_\Sigma$, we have $R_\Sigma^* \leq R_\Sigma$ and the modeled mismatch is ultimately bounded within the strategic reserve. This is the standard Khalil sector-Lyapunov ultimate-boundedness setup with the cross-term retained^[28] (Section 9.2; comparison-principle form). $\square$

Discrete-time form. For the discrete update $\delta_\Sigma^+ = \delta_\Sigma - \mathbf{F}(\delta_\Sigma) + \mathbf{w}$ with $\alpha_{ij}^{\text{ss}} \in [0, 1]$ and AM-GM with parameter $\eta = \mathcal{T}_\Sigma^{\text{bn,ss}} / (2(1 - \mathcal{T}_\Sigma^{\text{bn,ss}}))$, the steady-state bound is $\|\delta_\Sigma^*\| \leq 2\rho_\Sigma / \mathcal{T}_\Sigma^{\text{bn,ss}}$ — same scaling, factor-of-2 absorbable into the threshold constant.

Mean-square corollary. If instead $\mathbf{w}$ is interpreted as zero-mean stochastic disturbance with $\mathbb{E}[\|\mathbf{w}\|^2] \leq \rho_\Sigma^2$, the cross-term $\mathbb{E}[2\delta_\Sigma^\top \mathbf{w} \mid \delta_\Sigma] = 0$ vanishes in expectation, and the drift inequality gives mean-square ultimate boundedness $\sqrt{\mathbb{E}[V^*]} \leq \rho_\Sigma / \sqrt{2\mathcal{T}_\Sigma^{\text{bn,ss}}}$ under threshold $\mathcal{T}_\Sigma^{\text{bn,ss}} > \rho_\Sigma^2 / (2R_\Sigma^2)$. The deterministic-disturbance form above is the operative one for Theorem 4.1(A2) since non-stationarity-induced drift is structurally adversarial-aligned (consistently moving mismatch in one direction) rather than zero-mean.

990 **Sharpness** When the inequality reverses ($\mathcal{T}_{\Sigma}^{\text{bn,ss}} \leq \rho_{\Sigma}/R_{\Sigma}$), the adversarial disturbance $\mathbf{w}^*(t) =$
991 $\rho_{\Sigma} \delta_{\Sigma}(t)/\|\delta_{\Sigma}(t)\|$ saturates Cauchy–Schwarz and gives $d\|\delta_{\Sigma}\|/dt \geq -\mathcal{T}_{\Sigma}^{\text{bn,ss}} \|\delta_{\Sigma}\| + \rho_{\Sigma}$. By the
992 comparison principle, $\|\delta_{\Sigma}(t)\|$ exceeds any $R < \rho_{\Sigma}/\mathcal{T}_{\Sigma}^{\text{bn,ss}}$ in finite time. The bottleneck-element
993 refinement: pick $\mathbf{w}$ and δ_{Σ} both supported on $(i^*, j^*) \in \arg \min_{(i,j)} \alpha_{ij}^{\text{ss}}$ with $|w_{i^*j^*}| = \rho_{\Sigma}$ —
994 saturates the sector inequality on the contracting side and aligns with $\mathbf{w}$ on the disturbance side
995 simultaneously. The threshold is *sharp inside the diagonal sector model*. The theorem is silent about
996 non-diagonal correction architectures or stabilization mechanisms outside Model (Σ). $\square$

997 B.4 Proof of the impulsive ProST sector-level reduction

998 Within Model (Σ), idealize ProST^[8] as an impulsive system: between scheduled update times
999 $\{t_1, \dots, t_K\} \subset [0, T]$ the agent’s policy is held fixed and the strategic mismatch evolves under
1000 disturbance budget ρ_{Σ} alone (continuous-destabilizing in the Lyapunov $V = \|\delta_{\Sigma}\|^2$); at each update
1001 time t_i the policy is revised, contracting the modeled mismatch by per-update impulse gain $\gamma \in (0, 1]$
1002 so $\|\delta_{\Sigma}(t_i^+)\| \leq (1 - \gamma)\|\delta_{\Sigma}(t_i^-)\|$. Then $V(t_i^+) \leq (1 - \gamma)^2 V(t_i^-)$.

1003 This fits the Hespanha–Liberzon–Teel^[46] (Theorem 1) framework for impulsive systems with desta-
1004 bilizing continuous evolution and stabilizing impulses (the *reverse-ADT* branch). The continuous-
1005 evolution rate is $\lambda_c = \rho_{\Sigma}/R_{\Sigma}$ (from linearizing the disturbance contribution to $\dot{V} \approx +\lambda_c V + \gamma_c(\rho_{\Sigma})$).
1006 The impulse contraction in V is $\mu = (1 - \gamma)^2$. The reverse-ADT condition for ultimate boundedness —
1007 impulses cannot be too sparse — gives $\Delta_{\max} \cdot \lambda_c < -\ln \mu \iff \Delta_{\max} \cdot \rho_{\Sigma}/R_{\Sigma} < -\ln(1 - \gamma)^2$,
1008 where $\Delta_{\max} := \sup_i (t_{i+1} - t_i)$ is the longest inter-update gap. Under the condition the modeled
1009 mismatch is ultimately bounded within $R_{\Sigma}^* \approx \rho_{\Sigma} \Delta_{\max}/\gamma$ (leading order in γ).

1010 Under uniform blocks $\Delta_i = T/K$ the condition reduces to $K/T > \rho_{\Sigma}/(2\gamma R_{\Sigma})$ to leading order in γ
1011 — recovering ProST’s K/T form with the impulse gain made explicit. Under nonuniform schedules
1012 the impulsive form is *strictly stronger*: the longest block sets the threshold, not the average. The
1013 lemma is rigorous for the modeled impulsive sector dynamics; the per-update impulse gain γ is a
1014 domain quantity (depends on within-block sample size and update-time discount) and is left abstract.
1015 $\square$

1016 B.5 Proof of Key Lemma 3 (Loop generates interventional samples)

1017 By temporal ordering and the singular-trajectory commitment of Section 3, a_t causally precedes
1018 o_{t+1} . We invoke Pearl’s do-calculus^[35] (Theorem 3.4.1, Rule 2 — *action/observation exchange*):
1019 $P(o_{t+1} \mid \text{do}(a_t), H_t) = P(o_{t+1} \mid a_t, H_t)$ whenever $(o_{t+1} \perp\!\!\!\perp a_t \mid H_t)_{G_{\overline{a_t}}}$, where $G_{\overline{a_t}}$ denotes
1020 the graph G with all incoming arrows to a_t removed. In the singular-trajectory + agent-as-policy
1021 graph of Section 3, the only arrows into a_t come from H_t (the agent’s policy is $\pi_t(a_t \mid H_t)$); deleting
1022 them severs the only confounding path from a_t to o_{t+1} . Whether H_t d-separates a_t from o_{t+1} in $G_{\overline{a_t}}$
1023 is precisely the content of (C2). The identity $P(o_{t+1} \mid \text{do}(a_t), H_t) = P(o_{t+1} \mid a_t, H_t)$ holds on the
1024 support where $\pi_t(a_t \mid H_t) > 0$, which is automatic since a_t was drawn from π_t . $\square$

1025 *Remark on equivalent forms of (C2).* The d-separation condition $(o_{t+1} \perp\!\!\!\perp a_t \mid H_t)_{G_{\overline{a_t}}}$ is equivalent
1026 to the potential-outcome form $a_t \perp\!\!\!\perp Y^{(a_t)} \mid H_t$ (sequential ignorability in the Robins/Murphy sense),
1027 and to the truncated-factorization condition $P(\mathbf{U} \mid a_t, H_t) = P(\mathbf{U} \mid H_t)$ where $\mathbf{U}$ are the latent
1028 variables jointly affecting a_t and o_{t+1} — applied to the *prior* on $\mathbf{U}$ in the do-marginalization, not to
1029 the conditional on o_{t+1} . The three forms are interchangeable; the d-separation form is the most direct
1030 handle for the proof above.

1031 *On the role of (C1) and (C3).* The identification proof above is load-bearing on (C2) alone: (C1)’s full-
1032 support positivity and (C3)’s known-action-mechanism are *automatic* under the paper’s architecture —
1033 (C1) reduces to realized-action positivity ($\pi_t(a_t \mid H_t) > 0$ tautologically since a_t was drawn from π_t);
1034 (C3) reduces to agent-policy-queryability (Q_t is the agent’s own policy, internally accessible). Both
1035 are stated as named conditions for forward-compatibility with off-policy / counterfactual extensions
1036 where they would do real work; for the on-policy identification claim of this lemma, (C2) is the
1037 substantive condition.

1038 B.6 Proof of Key Lemma 4 (Bias bound)

1039 On the matching event $\{a_{\text{ag},t}^*(s) = \tilde{a}_t^*(s)\}$, the difference $\hat{D}_t(s) - D_t^{\text{true}}(s)$ vanishes identically (both
 1040 quantities equal $-\log Q_t$ at the same action). On the complementary misidentification event, the
 1041 support condition $Q_t(\cdot | s) \geq q_0$ at *both* $a_{\text{ag},t}^*(s)$ and $\tilde{a}_t^*(s)$ implies $\log Q_t(a_{\text{ag},t}^*(s) | s) \in [\log q_0, 0]$
 1042 and $\log Q_t(\tilde{a}_t^*(s) | s) \in [\log q_0, 0]$, so their absolute difference is at most $\log(1/q_0)$. Therefore
 1043 $|\hat{D}_t(s) - D_t^{\text{true}}(s)| \leq \mathbf{1}[a_{\text{ag},t}^*(s) \neq \tilde{a}_t^*(s)] \cdot \log(1/q_0)$. Taking expectations: $\mathbb{E}[|\hat{D}_t(s) - D_t^{\text{true}}(s)|] \leq$
 1044 $(1 - p_{\text{id}}(s)) \log(1/q_0)$ by definition of $p_{\text{id}}(s)$. $\square$

1045 **High-probability form** With probability $1 - \delta$ over a single round’s identification event, $|\hat{D}_t(s) -$
 1046 $D_t^{\text{true}}(s)| \leq \log(1/q_0) \cdot \mathbf{1}[\text{misdevent}]$ where $\Pr[\text{misdevent}] = 1 - p_{\text{id}}(s)$. The bias decomposes
 1047 cleanly: in Regime A ($p_{\text{id}}(s) \rightarrow 1$) it vanishes; in Regime B ($p_{\text{id}}(s) \in (0, 1)$) it is controlled by
 1048 misidentification mass; in Regime C ($p_{\text{id}}(s) \rightarrow 0$) it saturates at $\log(1/q_0)$. This proves Key Lemma
 1049 4. $\square$

1050 **Per-state, per-horizon-step lift (used in Theorem 4.1(v)).** The lemma applies pointwise at any
 1051 visited state s_h along a horizon- N_h trajectory: $\mathbb{E}[|\hat{D}_t(s_h) - D_t^{\text{true}}(s_h)|] \leq (1 - p_{\text{id}}(s_h)) \log(1/q_0)$,
 1052 with $p_{\text{id}}(s_h)$ the per-state argmax-correctness probability. Aggregating linearly over the horizon and
 1053 assuming the per-state floor $p_{\text{id}} := \min_s p_{\text{id}}(s)$, the per-round bias contribution to the trajectory-level
 1054 value gap is at most $N_h(1 - p_{\text{id}}) \log(1/q_0)$, contributing the second term of Theorem 4.1(v).

1055 **Remarks** - Both $a_{\text{ag},t}^*(s)$ and $\tilde{a}_t^*(s)$ must satisfy $Q_t \geq q_0$ for the $\log(1/q_0)$ ceiling to apply; the
 1056 support condition is on *both* points, not just the agent’s estimate. This is *strictly weaker* than the
 1057 perturbative identity’s full-support condition $Q(a) \geq q_0$ for *all* a (Appendix B.2) — the perturbative
 1058 identity needs full support because π_ϵ^* has positive mass on every action; the bias bound here needs
 1059 only two-point support. - This is *not* an estimation-from-samples lemma. If Q_t is unavailable internally
 1060 and must be estimated from on-policy rollouts, an additional concentration argument (Hoeffding
 1061 under iid fixed-policy samples, or martingale concentration for adaptive policies) is needed; in the
 1062 architectures this paper considers, Q_t is the agent’s policy and is internally available.

1063 C Auxiliary Mathematical Material

1064 This appendix collects supporting material used by the main results: convention hierarchy details, the
 1065 admissible-divergence family, direction-forcing argument, action-gap matching lower bound, tied-
 1066 optimum extensions, the structural-class theorem on gain-decay updates, strategic-tempo topologies,
 1067 deployment modes, and the chain-rule uniqueness theorem.

1068 C.1 Convention hierarchy: C1, C2, C3

1069 Three named continuation conventions form a hierarchy of increasing diagnostic power and computa-
 1070 tional cost.

1071 **C1 — One-step improvement (canonical default)** $\pi_{\text{cont}} = \pi_{\text{current}}$. Each action is evaluated
 1072 assuming current behavior continues afterward. No fixed-point computation, no global-optimality
 1073 assumption. Cheapest; weakest diagnostic.

1074 **C2 — Receding-horizon (N_r -step replanning)** At each future step, re-optimize
 1075 over a horizon of N_r steps using the model available at that step: $\pi_{\text{RH}}(M_\tau) =$
 1076 $\arg \max_\pi V_O(M_\tau, \pi; N_r)$ applied at each τ . Captures multi-step recovery: a goal that ap-
 1077 pears unattainable under frozen continuation may be reachable with replanning. Moderate cost;
 1078 moderate diagnostic.

1079 **C3 — Bellman** $\pi_{\text{cont}} = \pi^*$ where $\pi^* = \arg \max_\pi V_O(M_t, \pi; N_h)$. The continuation is the optimal
 1080 policy — a fixed-point equation. Strongest diagnostic; most expensive.

1081 **Proposition C.1 (Monotonicity).** *For any model M_t , horizon N_h , and policy class Π :*
 1082 $A_O^{(1)}(M_t; \Pi, N_h) \leq A_O^{\text{RH}}(M_t; \Pi, N_r, N_h) \leq A_O^{\text{B}}(M_t; \Pi, N_h).$

1083 *Proof.* C1 freezes continuation at π_{current} (possibly suboptimal); C2 re-optimizes periodically, so
 1084 $\pi_{\text{RH}} \succeq \pi_{\text{current}}$ at each future step; C3 uses the globally optimal π^* . A weakly better continuation
 1085 yields a weakly higher expected trajectory value; taking the supremum over the first action preserves
 1086 the ordering. $\square$

1087 **Corollary** $\delta_{\text{sat}}^{\text{B}} \leq \delta_{\text{sat}}^{\text{RH}} \leq \delta_{\text{sat}}^{(1)}$ (since $\delta_{\text{sat}} = V_O^{\min} - A_O$, higher A_O means lower δ_{sat}). And
 1088 $\delta_{\text{regret}}^{(1)} \leq \delta_{\text{regret}}^{\text{RH}} \leq \delta_{\text{regret}}^{\text{B}}$ (since $\delta_{\text{regret}} = A_O - V_O(M_t, \pi_{\text{current}}; N_h)$, higher A_O means higher δ_{regret}).
 1089 C1 is the most conservative diagnostic (most likely to diagnose “locally unattainable”); C3 is the most
 1090 accurate (least false “unattainable” diagnoses). The 2×2 diagnostic structure is preserved under all
 1091 three; only the inferential force varies.

1092 C.2 Admissible-divergence family for the regret bound

1093 The reverse-KL direction is forced by the deterministic- π^* regime (Appendix C.3). Within the
 1094 direction-forced family, multiple smooth f -divergences yield valid regret bounds:

Divergence	Bound on $R(Q)$	Tightness	Finite under det. π^* ?
$\text{TV}(\pi^*, Q)$	$V_{\max} \cdot \text{TV}$	Tight (extremal V)	Yes
$D_{\text{KL}}(\pi^* \ Q)$ via Pinsker	$V_{\max} \sqrt{D_{\text{KL}}/2}$	Loose by $\sqrt{\cdot}$	Yes
$D_{\text{KL}}(\pi^* \ Q)$ via point-mass identity	$V_{\max}(1 - e^{-D_{\text{KL}}})$	Tight (this paper)	Yes
$\chi^2(\pi^* \ Q)$ (Le Cam)	$V_{\max} \cdot \frac{1}{2} \sqrt{\chi^2}$	Looser than Pinsker	$\chi^2 = 1/Q(a^*) - 1$
$D_{\alpha}(\pi^* \ Q)$ (Rényi, $\alpha \geq 1$)	Various	Interpolates	Yes for $\alpha \geq 1$
$D_{\text{KL}}(Q \ \pi^*)$ (forward-KL)	$+\infty$	Vacuous	No

1096 Reverse-KL is selected uniquely within the direction-forced family by the chain-rule axiom (Ap-
 1097 pendix C.10). The point-mass identity supplies the *exact* bound on reverse-KL under determinis-
 1098 tic π^* ; the table reflects different bound shapes on the same divergence (with the BH inequality
 1099 $V_{\max} \sqrt{1 - e^{-D_{\text{KL}}}}$ as the looser general form to which the identity reduces outside the deterministic- π^*
 1100 corner).

1101 C.3 Direction-forcing argument

1102 For deterministic $\pi^* = \delta_{a^*}$ and any Q with $Q(a) > 0$ for some $a \neq a^*$: $D_{\text{KL}}(Q \| \pi^*) =$
 1103 $\sum_a Q(a) \log \frac{Q(a)}{\pi^*(a)} = \sum_{a \neq a^*} Q(a) \log \frac{Q(a)}{0} = +\infty$. A bound “ $R \leq +\infty$ ” is vacuous. The
 1104 reverse direction $D_{\text{KL}}(\pi^* \| Q)$ is finite (and equal to $-\log Q(a^*)$) whenever $Q(a^*) > 0$. The asym-
 1105 metry is forced by the regime: regret is a one-sided quantity (contributes only from Q ’s off-optimum
 1106 mass; π^* has no support off a^*); divergences whose role is to bound this one-sided quantity must
 1107 themselves be one-sided. Symmetric divergences (squared Hellinger, Jensen-Shannon, symmetrized
 1108 KL) introduce a vacuous symmetric term.

1109 C.4 Action-gap matching lower bound

1110 For any Q with $\Delta_{\min} = \min_{a \neq a^*} \Delta(a) > 0$: $R(Q) = \sum_{a \neq a^*} Q(a) \Delta(a) \geq \Delta_{\min} \sum_{a \neq a^*} Q(a) =$
 1111 $\Delta_{\min} \cdot (1 - Q(a^*)) = \Delta_{\min} \cdot \text{TV}(\pi^*, Q)$. By the point-mass identity (Lemma 5.1), $\text{TV}(\pi^*, Q) =$
 1112 $1 - e^{-D_{\text{KL}}(\pi^* \| Q)}$, giving the matching lower bound used in the two-sided characterization.

1113 The lower bound is tight when sub-optimal actions are uniformly bad ($\Delta_{\min} = \max_{a \neq a^*} \Delta(a)$). For
 1114 typical landscapes the gap between upper and lower bound is $V_{\max} - \Delta_{\min}$, controlled by the *spread*
 1115 of action gaps.

1116 C.5 Tied-optimum and softmax-smoothed extensions

1117 **Tied-optimum** If π^* has support on a tied-optimum set $\mathcal{A}^* = \{a : Q_O(a) = Q_O(a^*)\}$
 1118 with uniform mass, reverse-KL is finite whenever Q covers $\mathcal{A}^*$. The regret bound extends:
 1119 $R(Q) = \sum_{a \notin \mathcal{A}^*} Q(a) \Delta(a) \leq V_{\max} \cdot \mathbb{P}_Q(a \notin \mathcal{A}^*)$. A multi-action analog of the point-mass
 1120 identity holds with $\pi^*(a^*) = 1/|\mathcal{A}^*|$ replacing the point-mass form, yielding $D_{\text{KL}}(\pi^* \| Q) =$

1121 $-\sum_{a \in \mathcal{A}^*} (1/|\mathcal{A}^*|) \log(|\mathcal{A}^*|Q(a))$, a multi-action analog of $-\log Q(a^*)$. The two-sided bound be-
 1122 comes one-sided in general; the upper bound holds with the modified KL form.

1123 **Softmax-smoothed π^* with temperature $\tau \rightarrow 0$** Replace $\pi^* = \delta_{a^*}$ with $\pi_\tau^* \propto \exp(Q_O/\tau)$. As
 1124 $\tau \rightarrow 0$, $\pi_\tau^* \rightarrow \delta_{a^*}$. For finite τ , the *perturbative* identity (Appendix B.2) applies: $\text{TV}(\pi_\tau^*, Q) =$
 1125 $1 - e^{-D_{\text{KL}}(\pi_\tau^* \| Q)} + O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$ — exponentially small with polynomial prefactor. The
 1126 two-sided regret bound transfers with the same correction order. Outside the perturbative regime —
 1127 for genuinely high-entropy optima — the BH inequality $\text{TV} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$ becomes the relevant
 1128 general bound; Pinsker also applies, with a one-sided regret form.

1129 C.6 The gain-decay structural class $\mathcal{A}_{\text{decay}}$

1130 The persistence inequality $\mathcal{T}_{\Sigma}^{\text{bn,ss}} > \rho_{\Sigma}/R_{\Sigma}$ (Section 5.2) is an *instantaneous* check at the current
 1131 operating point. Define the structural class by gain-decay rather than sample-counting: $\mathcal{A}_{\text{decay}} :=$
 1132 $\{\text{updates whose effective sector gain } \alpha_{ij}^{(t)} \rightarrow 0 \text{ as } t \rightarrow \infty \text{ on every revisable element}\}$.

1133 **Theorem C.2** (Universal failure of $\mathcal{A}_{\text{decay}}$). *For any fixed $(\rho_{\Sigma}, R_{\Sigma})$ with $\rho_{\Sigma} > 0$, every agent in*
 1134 *$\mathcal{A}_{\text{decay}}$ eventually violates the persistence threshold $\mathcal{T}_{\Sigma}^{\text{bn,ss}} > \rho_{\Sigma}/R_{\Sigma}$.*

1135 *Proof.* At every element (i, j) , $\alpha_{ij}^{(t)} \rightarrow 0$ by definition of $\mathcal{A}_{\text{decay}}$, so $\min_{(i,j)} \alpha_{ij}^{(t)} \rightarrow 0$, so the
 1136 bottleneck $\mathcal{T}_{\Sigma}^{\text{bn}} \rightarrow 0$ with experience. Once $\mathcal{T}_{\Sigma}^{\text{bn,ss}} < \rho_{\Sigma}/R_{\Sigma}$, the threshold is violated. This holds
 1137 *regardless of prior calibration*: at any finite calibration level, the gain decays below threshold once
 1138 enough experience accumulates. $\square$

1139 *Scope clarification (per-element pull).* The class definition requires $\alpha_{ij}^{(t)} \rightarrow 0$ on *every* revisable
 1140 element — i.e., the agent is being pulled at every element repeatedly. Beta-Bernoulli with $\eta_{\text{edge},ij} =$
 1141 $1/(n_{ij} + 1)$ is in the class only for elements actually being updated; an agent that ignores a subset of
 1142 elements is outside the class for those elements (their gain remains at initial calibration), but inside
 1143 it for the elements it does touch. The theorem applies to the touched elements; persistence on the
 1144 untouched ones reduces to whatever calibration was set initially.

1145 This is a *structural* failure of the class, not a tuning problem. The class includes: - Count-accumulating
 1146 Bayesian updates without forgetting (e.g., Beta-Bernoulli with $\eta_{\text{edge},ij} = 1/(n_{ij} + 1)$ where $n_{ij} \rightarrow \infty$).
 1147 - Bounded-memory schemes with growing memory. - Observation-aggregating schemes without restart.
 1148 - Gradient-based methods with vanishing step sizes ($\eta_t = c/t^\beta$ for $\beta \in (0, 1]$, the Robbins–Monro
 1149 regime).

1150 Mechanisms *outside* $\mathcal{A}_{\text{decay}}$ — constant-step-size stochastic approximation, sliding-window updates,
 1151 bounded-memory learners, block-restart schemes, Kalman filters with bounded process noise —
 1152 maintain a finite gain ceiling and escape asymptotic decay, but then face a *bidirectional* threshold:

Table 1: Bidirectional thresholds for non-accumulating mechanisms outside $\mathcal{A}_{\text{decay}}$.

Mechanism	$\alpha_{\Sigma}^{\text{ss}}$	Prerequisite holds iff
Exponential forgetting with λ	$1 - \lambda$	$1 - \lambda > \rho_{\Sigma}/R_{\Sigma}$
Constant-step SA with rate η	η	$\eta > \rho_{\Sigma}/R_{\Sigma}$
Fixed-window mechanisms (sliding- window size W , bounded-memory size K , block-restart period T_R)	$1/W, 1/K, 1/T_R$	window/memory/period $< R_{\Sigma}/\rho_{\Sigma}$

1153 For each row, the threshold is *bidirectional* and sharp within the sector-Lyapunov reduction.

1154 C.7 Strategic-tempo across canonical topologies

1155 The bottleneck strategic tempo $\mathcal{T}_{\Sigma}^{\text{bn,ss}}$ of Lemma 5.2 is verified — and the resulting per-topology
 1156 threshold derived — across four canonical topologies under Beta-Bernoulli edge updates with per-
 1157 element forgetting:

- 1158 • **(T1) Single edge.** $\mathcal{T}_{\Sigma}^{\text{bn,ss}} = \nu \cdot \iota \cdot (1 - \lambda)$. All three factors load-bearing; setting any one to
1159 zero collapses the bottleneck.
- 1160 • **(T2) Two-edge AND chain, observable intermediate.** $\mathcal{T}_{\Sigma}^{\text{bn,ss}} = \min(\nu_1(1 - \lambda_1), \nu_1\theta_1(1 -$
1161 $\lambda_2))$. Edge 2’s effective observation rate is gated by edge 1’s success probability θ_1 —
1162 *depth-gated attenuation*. For depth- d chains, the bottleneck is $\min_k \prod_{j < k} \theta_j(1 - \lambda_k)$.
- 1163 • **(T3) Two-edge AND chain, unobservable intermediate.** Per-edge tempo ill-defined; plan-
1164 level bottleneck $\mathcal{T}_{\Sigma, \text{plan}}^{\text{bn,ss}} = \nu(1 - \lambda_{\Phi})$ over a single tracked plan-level quantity $\hat{\Phi} = p_1 p_2$.
- 1165 • **(T4) Two-arm OR node, ε -greedy.** $\mathcal{T}_{\Sigma}^{\text{bn,ss}} = \min((1 - \varepsilon)(1 - \lambda_1), \varepsilon(1 - \lambda_2))$. Action
1166 selection controls rate allocation; pure greedy ($\varepsilon = 0$) collapses the unexplored arm’s
1167 bottleneck — *exploration-gated*.

1168 The structural decomposition: AND-chains exhibit *depth-gated* geometric attenuation $\nu_k =$
1169 $\nu \prod_{j < k} \theta_j$; OR-nodes exhibit *exploration-gated* allocation. Mixed AND/OR DAGs interleave both.
1170 Each topology surfaces a different factor as the binding bottleneck.

1171 **Regime-C contamination is structurally fatal** $\iota_{ij} = 0$ at any single element $\Rightarrow \mathcal{T}_{\Sigma}^{\text{bn,ss}} = 0$. An
1172 agent with even one fully-confounded element cannot meet the threshold by forgetting alone.

1173 C.8 Three deployment modes of loop-Level-2

1174 The Pearl-do structure of closed-loop interventional access manifests at distinct layers with semanti-
1175 cally distinct interventional mechanisms:

- 1176 • **Mode 1 — agent-self-intervention.** The agent performs do-actions on its own action space
1177 as part of its ordinary adaptive loop. Intervention is on the agent’s own action; target is the
1178 environment’s response. This is Lemma 5.3’s primary content.
- 1179 • **Mode 2 — observer-on-sub-agent.** An observer external to a composite performs do-
1180 interventions on one sub-agent’s action space; target is another sub-agent’s response. Reveals
1181 cross-coupling structure that component-marginal observation cannot identify.
- 1182 • **Mode 3 — observer-on-agent-input** (sketched, future work). An observer intervenes on
1183 the agent’s observation channel; target is the agent’s subsequent policy. Useful for offline
1184 architectural-class diagnosis.

1185 Modes share the Pearl-do structure but differ in who intervenes on what. The unification is at the
1186 pattern level; the mechanism is semantically distinct per layer.

1187 C.9 Distinction from active inference and causal-RL precursors

1188 Action-perception-loop frameworks — active inference^[30,31], control-as-inference^[49], cybernet-
1189 ics^[50,51] — implicitly use the action-causes-observation observation. Our distinctive moves:

- 1190 • *Bareinboim-hierarchy connection.* Active inference / cybernetics rest on Bayesian-network
1191 (Level 1) generative models; we invoke the causal-hierarchy theorem^[29] to position the
1192 policy DAG as causal with do-conditioning in Q_O .
- 1193 • *Regime-indexed identifiability (A/B/C).* Active inference literature treats causal identifiability
1194 uniformly within modeling assumptions; we surface the regime split at framework level.
- 1195 • *Scope honesty.* We distinguish “data generated under intervention” from “cleanly identified
1196 do-estimates”;^[48] documents that the active-inference literature sometimes elides this.

1197 The causal-RL line^[13–18] is the direct ancestor for regime-indexed identifiability and on-policy inter-
1198 ventional access; all are stationary-MDP. Composition with non-stationarity is, to our knowledge,
1199 novel.

1200 C.10 Chain-rule uniqueness of reverse-KL

1201 **Theorem C.3** (Chain-rule uniqueness (Hobson 1969; Csiszár 1991)). *Let $D_f(P \parallel Q) =$*
 1202 *$\sum_x Q(x) f(P(x)/Q(x))$ be a smooth f -divergence with f convex and $f(1) = 0$. The chain rule*
 1203 *$D_f(P_{XY} \parallel Q_{XY}) = D_f(P_X \parallel Q_X) + \mathbb{E}_{P_X}[D_f(P_{Y|X} \parallel Q_{Y|X})]$ holds for all joint distributions*
 1204 *if and only if $f(t) = c \cdot t \log t$ for some $c > 0$ — i.e., D_f is reverse-KL up to positive scaling.*

1205 *Proof.* Take any joint P_{XY}, Q_{XY} on a finite product space. Writing $r_x := P(x)/Q(x)$ and $s_{y|x} :=$
 1206 $P(y|x)/Q(y|x)$, the joint ratio factorizes as $P_{XY}(x, y)/Q_{XY}(x, y) = r_x \cdot s_{y|x}$. Expanding both
 1207 sides of the chain rule: $D_f(P_{XY} \parallel Q_{XY}) = \sum_{x,y} Q_{XY}(x, y) f(r_x s_{y|x}) = \sum_x Q(x) \sum_y Q(y|x)$
 1208 $f(r_x s_{y|x})$, $D_f(P_X \parallel Q_X) + \mathbb{E}_{P_X}[D_f(P_{Y|X} \parallel Q_{Y|X})] = \sum_x Q(x) f(r_x) + \sum_x P(x) \sum_y Q(y|x)$
 1209 $f(s_{y|x})$, the second term using $\mathbb{E}_{P_X} = \sum_x P(x) = \sum_x Q(x) r_x$. Equating and collecting:
 1210 $\sum_x Q(x) \left[\sum_y Q(y|x) f(r_x s_{y|x}) - f(r_x) - r_x \sum_y Q(y|x) f(s_{y|x}) \right] = 0$. The chain rule
 1211 must hold for *all* joints, including those where r_x takes arbitrary positive value at one x and $s_{y|x}$
 1212 varies independently — which forces the bracketed expression to vanish pointwise. A two-point
 1213 reduction (take Y binary with $s_{y|x}$ taking values s and $1/s$ on equal-weight $Q(y|x)$) collapses
 1214 the inner sums and yields the functional equation $f(rs) = f(r) + r f(s)$ for all $r, s > 0$.
 1215 With f convex and $f(1) = 0$, the unique solution is $f(t) = c \cdot t \log t$ for $c > 0$ ^[52] (§4) — this
 1216 gives $D_f(P \parallel Q) = c \sum_x Q(x) (P(x)/Q(x)) \log(P(x)/Q(x)) = c \sum_x P(x) \log(P(x)/Q(x)) =$
 1217 $c \cdot D_{\text{KL}}(P \parallel Q)$, reverse-KL up to positive scaling. $\square$

1218 **References** Hobson 1969 (“A new theorem of information theory,” *J. Stat. Phys.*); Csiszár 1991
 1219 (“Why least squares and maximum entropy?” *Annals of Statistics*; Theorem 3 corollary, Theorem 5);
 1220 Shore-Johnson 1980 (“Axiomatic derivation of the principle of maximum entropy,” *IEEE Trans. Info.*
 1221 *Theory*, system-independence axiom); Sanov 1957 (large-deviation rate function); Aczél-Daróczy
 1222 1975 (functional-equation machinery).

1223 These references give *structurally equivalent reformulations* of the same axiom. The Cauchy functional
 1224 equation each reduces to is the common content. No known uniqueness route outside the independence-
 1225 on-sub-problems family exists.

1226 **Why other family members fail the chain rule** Concrete counterexample for χ^2 : take Q_X uniform
 1227 on $\{x_1, x_2\}$, $P_X = (3/4, 1/4)$, $Q(y|x)$ uniform, $P(y|x) = (3/4, 1/4)$. Direct calculation gives
 1228 $\chi^2(P_{XY} \parallel Q_{XY}) = 9/16$ while $\chi^2(P_X \parallel Q_X) + \mathbb{E}_{P_X}[\chi^2(P_{Y|X} \parallel Q_{Y|X})] = 1/4 + 1/4 = 8/16$.
 1229 Non-additive. Rényi- α for $\alpha \neq 1$ fails analogously; squared Hellinger likewise fails.

1230 D Algorithm Sketch and Base-Learner Instantiation

1231 A practical algorithm achieving the joint guarantees of Theorem 4.1 requires four components, each
 1232 translating one of the assumptions (A1)–(A5) into operational form.

1233 **(a) Identity-tight base learner satisfying (A5)** In the stationary deterministic- π^* bandit case ($N_h =$
 1234 1), Thompson sampling^[19] (posterior-randomized; pointwise $\hat{Q}_t(a^*) > 0$) and vanilla deterministic
 1235 UCB^[27] (Dirac-deployed; $\hat{Q}_t(a^*) \in \{0, 1\}$, with the (A1) extended-real reading of Theorem 4.1) both
 1236 achieve the per-round coordinate in expectation: $\mathbb{E}[1 - e^{-K_t}] = \mathbb{E}[1 - \hat{Q}_t(a^*)] = O\left(\frac{\log t}{t \Delta_{\min}^2}\right)$,
 1237 following from Lattimore and Szepesvári 2020^[27] Theorem 7.1 ($\mathbb{E}[N_a(t)] \leq 16 \log(t)/\Delta_a^2 + O(1)$
 1238 per suboptimal arm; aggregating over arms with the per-state floor gives the per-round form). This
 1239 satisfies (A5) of Theorem 4.1 with the *stronger logarithmic* rate, yielding cumulative dynamic
 1240 regret $O(V_{\max}(B_T + 1) \log(T/(B_T + 1))/\Delta_{\min}^2)$. The $V_{\max} \cdot \text{TV}$ chain is structurally one factor of
 1241 $\Delta_{\min}$ looser than direct gap-aware UCB analysis (which gives $O(\log T/\Delta_{\min})$ cumulative regret per
 1242 block by weighting each suboptimal pull by its own $\Delta(a)$ rather than $V_{\max}$); this is a feature of the
 1243 unification — the framework’s contribution is composition + non-stationarity-handling + identity-
 1244 coordinate exactness, not a sharper bandit constant. For $N_h > 1$ MDPs, UCRL2^[53] and UCBVI^[43]
 1245 give per-block cumulative trajectory-level regret $\tilde{O}(N_h^{3/2} \sqrt{SAL})$ with S, A the state/action sizes;
 1246 rewriting in the occupancy-weighted coordinate satisfies (A5) with c of order $\tilde{O}(\sqrt{N_h SA})$, lifting

cleanly to the $\tilde{O}(N_h^2 \sqrt{SA(B_T + 1)T})$ cumulative rate — same piecewise-stationary B_T family as restart-on-change variants of^[1] (the continuous-variation regime of^[3] is structurally distinct; see Section 4 *Comparator regime*). The occupancy-weighted form is the natural object for trajectory-level base learners; the per-state KL/TV identity is preserved underneath. Information-directed sampling^[20] requires a different analysis here because $H(\pi^*) = 0$ collapses its information ratio at the deterministic- π^* corner; we leave the IDS analysis as future work.

(b) Multi-factor forgetting schedule satisfying (A2) Choose per-element discount rates $\{\lambda_{ij}\}$ such that the bottleneck $\mathcal{T}_{\Sigma}^{\text{bn,ss}} = \min_{(i,j)} \nu_{ij} \ell_{ij} (1 - \lambda_{ij})$ exceeds ρ_{Σ}/R_{Σ} . When ρ_{Σ} is unknown, the^[2] black-box reduction gives an adaptive choice (applied per element if heterogeneous); when ρ_{Σ} is known via a variation-budget^[1] or change-point detection^[39], the choice is direct. As a fail-fast pre-check, $\mathcal{T}_{\Sigma}^{\text{agg,ss}} > |E| \cdot \rho_{\Sigma}/R_{\Sigma}$ is necessary.

(c) Identifiability check satisfying (A3) Test whether the loop data is in Regime A (full identifiability), Regime B (partial), or Regime C (none). In Regime A, no adjustment is needed. In Regime B, apply^[17] (DOVI)-style confounding-bias adjustment. In Regime C, the bound is provable but not realizable; algorithm flags the regime as out-of-scope.

(d) Two-gap diagnostic readout satisfying (A4) Compute δ_{sat} (requires A_O , intractable in general; estimable via the policy-class supremum over recent rounds) and δ_{regret} (requires V_O at current policy; tractable in simulation). Route to corrective action class.

The algorithm is sketch-grade; full instantiation and empirical evaluation are deferred to a follow-up paper.

E Pinsker vs. Point-Mass Identity Numerical Comparison

We compare $V_{\max}(1 - e^{-D_{\text{KL}}})$ (point-mass identity bound, exact under deterministic π^* per Lemma 5.1 and the two-sided regret theorem) with $V_{\max} \sqrt{D_{\text{KL}}}/2$ (Pinsker bound, also valid under deterministic π^* but loose), $V_{\max} \sqrt{1 - e^{-D_{\text{KL}}}}$ (Bretagnolle–Huber inequality, the general envelope below which the identity sits at this corner), and the trivial envelope $V_{\max}$. Set $V_{\max} = 1$ for normalization.

Table 2: Identity vs. Pinsker vs. BH at $V_{\max} = 1$.

D_{KL}	$1 - e^{-D_{\text{KL}}}$ (identity)	$\sqrt{1 - e^{-D_{\text{KL}}}}$ (BH)	$\sqrt{D_{\text{KL}}}/2$ (Pinsker)	ratio
0.01	0.00995	0.0998	0.0707	7.10×
0.1	0.0952	0.308	0.224	2.35×
0.5	0.393	0.627	0.500	1.27×
1.0	0.632	0.795	0.707	1.12×
2.0	0.865	0.930	1.000	1.16×
4.0	0.982	0.991	1.414	1.02×
10.0	0.99995	0.99998	2.236	1.00×

The point-mass identity bound is uniformly tighter than Pinsker, by a factor of $7 \times$ at $D_{\text{KL}} = 0.01$ and converging to 1 as D_{KL} grows large (where both saturate at $V_{\max}$). It is also uniformly tighter than the BH inequality $\sqrt{1 - e^{-D_{\text{KL}}}}$ — the relation $x < \sqrt{x}$ on $(0, 1)$ — although both saturate together at $V_{\max}$ for large D_{KL} . Pinsker, by contrast, hits the trivial envelope $V_{\max} = 1$ at $D_{\text{KL}} = 2$ and exceeds it beyond — vacuous from $D_{\text{KL}} = 4$ onward (where the unclipped form returns 1.414), fully vacuous by $D_{\text{KL}} = 10$.

The matching lower bound $\Delta_{\min}(1 - e^{-D_{\text{KL}}})$ has the same shape, scaled by $\Delta_{\min}/V_{\max}$. In the extremal value landscape ($\Delta_{\min} = V_{\max}$), the upper and lower bounds coincide and the regret is *identified exactly* by the KL coordinate. For typical landscapes, regret and KL are Lipschitz-equivalent with constants $\Delta_{\min}/V_{\max}$ and 1.

1282 F Prior-Art Retrieval Summary

1283 We conducted a structured prior-art retrieval (via Undermind) with the goal of identifying any frame-
1284 work composing all four target elements: (1) goal-feasibility-vs-policy-quality two-gap diagnostic;
1285 (2) an exact reverse-KL/TV regret identity (point-mass) under deterministic- π^* , or — more loosely
1286 — Bretagnolle–Huber-style use in regret analysis; (3) strategic-tempo / forgetting prerequisite tying
1287 convergence to rate of useful policy revision; (4) closed-loop interventional access making regret
1288 bounds learnable from on-policy data.

1289 **Result: 63 papers retrieved, abstract-level coverage estimated 75%, no direct anticipation in**
1290 **the retrieved set** The landscape splits into four largely separate strands corresponding to our four
1291 components; we did not find a published framework composing all four in this retrieval. Retrieval
1292 queries used the structured prior-art tool Undermind; the four-strand cite-and-distinguish in Section 2
1293 surfaces the closest neighbor in each strand.

1294 **Strongest negative signal (in the retrieved corpus): the Bretagnolle–Huber inequality is absent**
1295 **as an upper-bound coordinate for regret analysis in the retrieved RL / non-stationary literature**
1296 The information-theoretic regret line —^[19–24] — uses entropy / mutual information / information
1297 ratio / Pinsker / Hellinger; we did not find the BH inequality deployed *as an upper-bound regret*
1298 *coordinate*. (BH is well-known in bandit *lower-bound* proofs via Le Cam’s method and Tsybakov’s
1299 framework^[26], including in standard RL textbook treatments^[27]; that use is not contested here.) The
1300 *exact point-mass identity at the deterministic- π corner** (Lemma 5.1) — which strictly improves both
1301 Pinsker and the BH envelope at this corner — was likewise not found in the retrieved corpus, in either
1302 upper-bound or lower-bound use. We frame this as a literature-search observation against an explicitly
1303 bounded retrieval (63 papers, ~75% abstract-level coverage), not a formal absence claim against all
1304 published RL work; reviewers should read it as “to our knowledge / in the retrieved set” rather than
1305 “absolutely novel.”

1306 **Closest neighbors per strand:**^[1] (Strand 1 — variation-budget dynamic regret);^[5] (Strand 2 —
1307 closest two-term decomposition, different axis);^[8] (Strand 3 — closest tempo result, different form);^[14]
1308 (Strand 4 — closest causal-RL, stationary only).

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1315 four-component composition (two-gap diagnostic, point-mass reverse-KL/TV identity, multi-
1316 factor strategic tempo with forgetting prerequisite, closed-loop interventional access) and
1317 the cumulative dynamic regret rate. Each component is developed in the main-result and
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